

APIx

The Real-time Airfare Price Index for India

An automated, ethically-sourced, CPI-2024-native price statistics platform for the augmentation of India’s Consumer Price Index — with a formal index-number specification, a dual-time-basis methodology, and a zero-circumvention collection architecture.

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Title	Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI)
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Theme	Travel & Tourism
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How to read this document.

- Part 1** Executive summary and the central reframe.
- Part 2** What the problem actually is in August 2026 — not what the PS background says.
- Part 3** Index-number theory: elementary formulae, chain drift, multilateral methods.
- Part 4** The formal APIx specification: product key, aggregation stages, availability adjustment, elasticity.
- Part 5** Data: sources, URLs, route basket design, schema, quality contract.
- Part 6** The system: architecture, ethical collection framework, ingestion, storage, API, dashboard.
- Part 7** Ten USPs, each with a demo path, a build cost, and the hostile question it must survive.
- Part 8** Delivery: the 30-day back-test, team split, risk register, judge Q&A drill.
- Part 9** Sourced reference list with live URLs.

Compiled August 2026. All external facts are footnoted or listed in Part 9 with live URLs. Figures marked [VERIFY] are values that change and must be re-pulled from the primary source on the day of the build — do not quote them from this document on stage.

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PART 1

Executive Summary

1.1 The one-paragraph version

APIx is a production-grade price statistics platform that collects domestic Indian airfare quotes at high frequency from a legally defensible ladder of sources, cleans and decomposes them into base fare, statutory levies, taxes and platform charges, and compiles them into a family of published index numbers that are *directly consumable by MoSPI's Price Statistics Division and by the RBI's Monetary Policy Department*. The index is constructed on the same mathematical machinery that MoSPI itself adopted for the CPI 2024 series — Jevons at the elementary level, Young/Modified Laspeyres above it — so that it can be plugged into COICOP sub-class 07.3.3 (*Passenger transport by air*) without a methodological argument. Everything above that baseline — daily frequency, route granularity, lead-time resolution, availability adjustment, and a nowcast of the Transport division — is capability the official series does not currently have and cannot cheaply acquire.

1.2 The reframe: read the problem statement against reality

The single most important thing in this document.

The PS background says CPI air fares are collected “primarily through manual price-collection from a limited set of outlets and ticketing offices.” **That statement describes the CPI 2012 series, which was retired on 12 February 2026.** Under the CPI 2024 series — released on 12 February 2026 with base 2024=100 — MoSPI's own FAQ states plainly that “*Airfares are collected through well-known online platforms*”, that online prices are collected *weekly*, and that twelve online markets have been added across twelve towns with population above 25 lakh.^a

^aMoSPI, *Frequently Asked Questions on CPI 2024 Series*, Annexure V to the CPI press release of 12 February 2026, Q.14, Q.8 and Q.27.

Most teams will not read the FAQ. They will build a scraper, put a Chart.js line on a dashboard, and pitch it as “solving” manual collection. A MoSPI DIID evaluator knows their own methodology. The moment a team says “currently MoSPI collects airfares manually,” the evaluator stops listening.

So the honest framing — and the one that wins — is a *delta* argument. What does the CPI 2024 airfare item index still not do?

Dimension	CPI 2024 airfare item	What APIx adds
Frequency	Weekly collection, monthly publication, 12-day lag	Daily index, published T+1; monthly figure available before the official release
Geography	Twelve online markets tied to towns >25 lakh population	Route-level (city-pair) index over a DGCA-traffic-weighted basket; sector heat map
Lead time	Not resolved as a published dimension	Five advance-purchase windows (T+1, +7, +15, +30, +45) with a full elasticity surface
Availability	Sold-out flights are unit non-response	Explicit availability adjustment; index does not silently drop when cheap buckets vanish
Time basis	Single basis	Dual basis: <i>booking-date</i> (what a consumer pays today) and <i>travel-date</i> (CPI-consistent, ONS-style)
Decomposition	Total fare	Base fare / YQ / UDF / ASF / RCS levy / GST / OTA convenience fee, indexed separately
Consumers	MoSPI internal	MoSPI PSD, RBI MPD, DGCA TMU, MoCA, and the public — one SDMX-shaped API

1.3 The second reframe: the PS contains an internal contradiction

The PS asks the system to handle “dynamic CAPTCHAs, anti-bot measures, IP rotation, and session management” and, in the same sentence, to remain “compliant with the robots.txt and terms of service of source websites.” These two clauses cannot both be satisfied. Defeating a CAPTCHA is by definition circumventing a technical access control; every major Indian OTA’s terms of use prohibit automated access; and under Section 43 read with Section 66 of the Information Technology Act, 2000, extracting data from a computer resource without the owner’s permission attracts a compensation-and-penalty mechanism. A prototype that demos a CAPTCHA solver is demonstrating a liability, not a capability.

APIx resolves the contradiction by **inverting the priority**: compliance is the hard constraint, coverage is the optimisation target. We implement a four-rung *Source Ladder* (Part IV, §6.2.2) that exhausts licensed and publicly-sanctioned channels before it ever touches an HTML page, records the provenance and legal basis of every single quote in an append-only ledger, and treats a CAPTCHA appearing on a page as a *stop signal that is logged as a governance event*, not an obstacle to route around. This is not a weaker system. For a Government of India statistical product it is the only shippable one, and it is a USP no scraping-first team can retrofit in 36 hours.

1.4 What gets built

- Collection engine** — Python; `httpx/Scrapy` for API and static sources, `Playwright` for JS-rendered pages that permit automated access, a robots.txt-aware politeness governor, per-host token buckets, and a circuit breaker that halts a host on the first hard block. Scheduled by `APScheduler/Celery beat`; containerised.
- Quote store** — PostgreSQL 16 with TimescaleDB hypertables for the quote time series, plus a Parquet cold tier queried by DuckDB. Star schema: one immutable `fare_quote` fact, six conformed dimensions, full slowly-changing history on airport charges and tax rates.
- Index engine** — a pure, side-effect-free Python module that takes a set of cleaned quotes and a weight vector and emits index numbers. Jevons elementary aggregates; Young/Modified Laspeyres aggregation; GEKS-Jevons and Time-Product-Dummy implementations as robustness

variants; bootstrap confidence intervals; deterministic and unit-tested against worked examples from the IMF CPI Manual.

- d. **Dashboard** — React + Recharts. Daily/weekly/monthly APIx; sector heat map; lead-time elasticity surface; fare-component stack; availability gauge; back-test overlay against DGCA and CPI series.
- e. **Public API** — FastAPI, OpenAPI 3.1, JSON and SDMX-JSON output, API-key auth, documented rate limits, and a `/provenance` endpoint that returns the audit chain behind any published number.
- f. **Evidence pack** — 30 days of back-tested results, a reproducibility script that regenerates every published figure from raw quotes, automated test suite with coverage report, and a written methodology note in the format MoSPI uses for its own discussion papers.

1.5 The ten USPs, in one line each

#	USP	The claim
1	Zero-Circumvention Architecture	The only prototype that can be deployed by a Government ministry on day one without a legal review killing it.
2	CPI-2024-Native by construction	Same formulae, same COICOP code, same linking arithmetic as MoSPI — integration is configuration, not migration.
3	Dual-Time-Basis Index	Publishes both the acquisition-basis and the use-basis index, and reconciles them. No Indian series does this.
4	Availability-Adjusted Index	Sold-out cheap fares are treated as a price signal, not as missing data. Fixes a genuine downward bias.
5	Lead-Time Elasticity Surface	The fare-vs-lead-time surface as a first-class published product with estimated elasticities.
6	Rigorous Back-Test Harness	Formal agreement statistics against DGCA TMU movements and the CPI air fare item, not an eyeballed overlay.
7	CPI Transport Nowcast	APIx as a leading indicator: a documented lead-lag analysis and a nowcast of division 07.
8	Hash-Chained Reproducibility	Every published number regenerable and tamper-evident. Official-statistics grade.
9	Frugal Deployment Envelope	Costed to run on NIC/MeghRaj-class infrastructure; no commercial proxy dependency.
10	Fare Transparency Layer	Directly implements the Parliamentary Standing Committee’s “Pricing Transparency Index” recommendation.

1.6 Brutal self-assessment

Where this project is genuinely hard, stated up front:

- **The booking lead-time distribution for India is not public.** The weights ω_τ over advance-purchase windows are the single largest judgement call in the index. We do not have a DGCA or MoSPI source for it. We handle this by making ω_τ an explicit, documented, user-tunable parameter and shipping a sensitivity analysis; we do *not* pretend it is measured. Judges will ask. Say this.
- **Thirty days of back-test cannot be faked.** If collection starts on hackathon day, there is no history. The mitigation is a two-track back-test: a live 30-day run started as early as possible, plus a synthetic replay harness validated against published DGCA monthly average-fare movements. Be honest about which chart is which.
- **Coverage under the compliance constraint will be lower than a scrape-everything team’s.** We will have fewer carriers or fewer OTAs. The answer is not to hide it; it is to show the *coverage ledger*, quantify the resulting index variance, and argue that a defensible 60% is worth more to a ministry than an inadmissible 95%.

- **Airfare is a small CPI weight.** Transport is 8.796% of combined CPI 2024; air fare is a fraction of that. Do not oversell APIx as moving headline inflation. Sell it as (a) methodological infrastructure MoSPI can reuse for hotels, cabs, and e-commerce, and (b) a consumer-transparency and regulatory instrument for MoCA/DGCA.

PART 2

Problem Understanding and Positioning

2.1 What the CPI is, precisely

The Consumer Price Index measures the change over time in the general level of retail prices of a fixed basket of goods and services that households buy for consumption. It is India's headline retail inflation measure and the operational target of the RBI's flexible inflation targeting framework. Understanding four technical terms is a prerequisite for everything in Part II.

Term	Meaning, and its value in CPI 2024
Index reference period	The period for which the index equals 100. For the current series: calendar year 2024 = 100 .
Weight reference period	The period from which expenditure weights are derived. For CPI 2024: HCES 2023–24 , using the Mixed Modified Reference Period.
Price reference period	The period during which base prices are collected. For CPI 2024: calendar year 2024 .
Elementary aggregate	The lowest level at which an index is computed, and the only level at which no expenditure weights are available. This is where the choice of Jevons vs. Dutot vs. Carli lives, and it is where APIx does most of its work.

2.1.1 The CPI 2024 series: the facts you must have straight

Attribute	CPI 2024 (base 2024=100)
First release	12 February 2026, for January 2026 (2.75% y-o-y, provisional)
Predecessor	CPI 2012 series, last released for December 2025
Weights source	HCES 2023–24 (MMRP)
Classification	COICOP 2018 (UN Statistics Division): 12 Divisions, 43 Groups, 92 Classes, 162 Sub-classes
Weighted items	358 at All-India level (308 goods, 50 services), up from 299
Sample	1,465 rural markets, 1,395 urban markets across 434 towns, plus 12 online markets
Collection	Tablet-based CAPI by the Field Operations Division of NSO; online prices weekly
Elementary formula	Jevons
Higher-level formula	Young / Modified Laspeyres
Airfare collection	“through well-known online platforms”
Rail fare	Administrative data from the Ministry of Railways
Fuel	Retail selling prices from PPAC, Ministry of Petroleum & Natural Gas
Dissemination	e-Sankhyiki portal; monthly indices for Divisions, Groups, Classes, Sub-classes and items, All-India and State/UT
Back series	Jan 2013 – Dec 2024, computed with a linking factor over overlap year 2025
Linking factors (General)	Rural 0.5222, Urban 0.5320, Combined 0.5267

Source: MoSPI, CPI 2024 FAQ (Annexure V, 12 Feb 2026); MoSPI CPI press releases Jan–Jun 2026; MoSPI Expert Group Report on Comprehensive Updation of CPI.

2.1.2 Where air fare sits in the tree

Under COICOP 2018 the relevant path is:

Level	Code	Label
Division	07	Transport
Group	07.3	Transport services
Class	07.3.3	Passenger transport by air
Sub-class	07.3.3.x	Domestic / international air passenger transport

Division-level weights in CPI 2024 are published:

Division	Rural		Urban		Combined	
	2012	2024	2012	2024	2012	2024
Food and beverages	50.922	41.983	32.811	30.251	42.617	36.753
Housing, water, electricity, gas	7.983	11.764	27.294	25.000	16.888	17.665
Transport (07)	5.645	8.644	7.129	8.985	6.394	8.796
Information and communication	2.818	3.647	3.906	3.563	3.323	3.609
Health	6.839	6.764	4.820	5.275	5.900	6.100

Source: MoSPI CPI 2024 FAQ, Q.39. Abridged — the full 12-division table is in the source.

[VERIFY] before the pitch. The *item-level* weight for passenger transport by air is published in the item-wise weight annexure at <https://cpi.mospi.gov.in> (Announcements tab) and on <https://mospi.gov.in>. Pull the exact number the week of the hackathon and quote it. Two things to note when you do: (i) the Transport division rose from 6.394 to 8.796 combined — a 38% relative increase in weight, which is a strong “this matters more now” talking point; (ii) air fare is only a slice of 07.3, so do not overstate the headline impact. Honest framing beats a big number.

2.2 The institutional map: who consumes what

A pitch that names the right desks is worth more than one that says “the government”.

Body	Unit	Relationship to APIx
MoSPI / NSO	Price Statistics Division (PSD)	Owns CPI methodology, weights, item basket, COICOP mapping. This is who the PS means by “PSD” in “index-construction module based on PSD given routes and weights.” PSD is the consumer of the index engine and the supplier of the weight vector.
MoSPI / NSO	Field Operations Division (FOD)	Runs the actual price collection, CAPI tablets, 434 towns. APIx does not replace FOD; it supplements the online-market stream.
MoSPI	Data Informatics & Innovation Division (DIID)	The PS owner. Mandate is alternative data sources and modernisation. Speak to DIID’s incentives: reusable pipeline, not a one-off scraper.
RBI	Monetary Policy Department	Consumes CPI for the flexible inflation targeting framework. Values <i>timeliness</i> above all — a credible T+1 now-cast of a division is directly useful.
MoCA / DGCA	Tariff Monitoring Unit (TMU)	Established 2010. Monitors fares on 78 selected routes, monthly, manually, by looking at airline websites , covering roughly 27% of domestic traffic . This is the unit whose work APIx most directly automates.
AERA	Tariff orders	Sets User Development Fee per airport. Source of truth for the UDF component of fare decomposition.
BCAS / MoCA	ASF notifications	Aviation Security Fee. Second statutory levy component.

Source: Rajya Sabha reply by MoS Civil Aviation, 8 Dec 2025 (TMU, 78 routes, ≈27% of traffic); MoSPI Expert Group Report (PSD role in COICOP mapping).

2.2.1 The DGCA TMU is the strongest single hook in this PS

Three facts, in sequence, are the most persuasive thirty seconds you have:

1. The DGCA's Tariff Monitoring Unit monitors **78 routes, manually, once a month, by opening airline websites** — covering about 27% of domestic traffic.
2. In its 375th Report (*Demands for Grants 2025–26, Ministry of Civil Aviation*), the Parliamentary Standing Committee on Transport, Tourism and Culture found that the DGCA *lacks capacity to proactively monitor airfare trends*, and recommended an AI-driven predictive monitoring system, a public **“Pricing Transparency Index”**, and an **“Airfare Vigil”** reporting app.
3. In July 2026 the Ministry told Parliament that average airfare on **72 domestic sectors rose 20.5%** comparing June 2026 with March 2025, attributing it to ATF — which is 35–40% of airline operating cost and which spiked over 100% on the West Asia conflict.

APIx is, in effect, a working prototype of two out of three of those Parliamentary recommendations, built for the statistics ministry rather than the aviation ministry. That is a genuinely strong positioning line and almost no competing team will find it.

2.3 Why airfare is the hardest item in the CPI basket

Pathology	Why it breaks naive price collection
Dynamic pricing	The same seat on the same flight has a different price every hour. There is no “the price” to observe — only a price <i>conditional on when you look and how far ahead you are looking</i> .
Perishable inventory	A seat unsold at departure is worthless. Revenue management therefore raises price as inventory depletes and as departure approaches — and sometimes <i>cuts</i> it to dump distressed inventory. The lead-time–price relation is not monotone.
Bucketing	Airlines sell from discrete fare buckets (RBDs). Observed prices are a step function, not a continuum, so the empirical distribution is multi-modal. Mean-based statistics are badly behaved; log-space and rank-based statistics are not.
Availability censoring	When the cheap buckets sell out you observe only expensive quotes. If you drop the missing cheap ones as “no data”, the index is <i>biased downward</i> in exactly the periods you care about (festivals, disruptions). This is the classic selection-on-the-dependent-variable trap.
Bundle drift	“Base fare” means different things across carriers: cabin bag only, checked bag included, seat selection, meal, change fee. Comparing an IndiGo <i>Saver</i> with an Air India <i>Comfort</i> is a quality comparison, not a price comparison.
Personalisation	Prices may vary by cookie state, device, geography and account. An unauthenticated, cookie-clean, India-egress collection is a <i>methodological requirement</i> , not a convenience.
Route entry and exit	Carriers add and drop sectors constantly. A fixed matched sample decays; an unmatched sample chain-drifts. This is precisely the problem multilateral index methods were invented to solve.
Acquisition vs. use	Do you record the price when the ticket is bought or when the flight is taken? The two give different indices. ONS records the change in the month the flight departs, not when the ticket is bought . Most teams will not even know this is a question.

2.4 The Indian market, in numbers you can quote

Fact	Source / date
Domestic airlines carried 864.04 lakh passengers in Jan–Jun 2026, +1.44% y-o-y	DGCA traffic report, June 2026
July 2026: $\approx$ 1.20 crore domestic passengers, –4.8% y-o-y; Jan–Jul 2026 total 984.03 lakh	DGCA, Aug 2026
IndiGo domestic market share 67.4% (July 2026); record 66.3% in June 2026	DGCA monthly
Air India Group $\approx$ 24%; Akasa 5.5%; SpiceJet 1.6% (July 2026)	DGCA monthly
Passenger load factors June 2026: Akasa 92.2%, SpiceJet 87.8%, AI Group 85.5%, IndiGo 85.1%	DGCA monthly
Cancellation rate 0.63% (June 2026); technical 39.6%, operational 36.3%, weather 20.3%	DGCA monthly
ATF is 35–40% of airline operating cost ; ATF spiked $>100\%$ on the West Asia conflict	MoCA reply to Parliament, July 2026
Average fare on 72 domestic sectors up 20.5% , June 2026 vs. March 2025	MoCA reply to Parliament, July 2026
DGCA TMU monitors 78 routes manually, monthly, $\approx$ 27% of domestic traffic	Rajya Sabha, 8 Dec 2025
India: third-largest civil aviation market, $\approx$ 4.2% of global air traffic (2024)	IATA / industry

Market-structure consequence for the index design. With one carrier at $\approx$ 67% share, a carrier-equal-weighted index is **wrong**. It would let SpiceJet’s 1.6%-share pricing move the national index as much as IndiGo’s 67%. APIx therefore weights carriers *within* each route by that carrier’s share of passengers on that route, not by national share and not equally. This is a detail that separates a statistics project from a dashboard project — see §4.4.

2.5 Gap analysis: what genuinely does not exist today

Capability	CPI 2024	DGCA TMU	APIx
Automated collection	Partially	No (manual)	Yes
Daily frequency	No (weekly collect)	No (monthly)	Yes
Route-level index	No	Fare levels only	Yes
Advance-purchase dimension	No	No	Yes, 5 windows
Availability adjustment	No	No	Yes
Fare-component decomposition	No	No	Yes
Published elasticity surface	No	No	Yes
Machine-readable public API	Portal only	No	Yes (SDMX-JSON)
Provenance audit trail	Internal	Internal	Public endpoint
Nowcast of Division 07	No	No	Yes

The honest conclusion: **APIx is not filling a hole in CPI; it is building the instrument layer that neither the statistics ministry nor the aviation regulator currently has, and handing the same instrument to both.** That is the sentence to put on the final slide.

PART 3

The Statistical Theory

This part is the reason the project is a *statistics* project and not a scraping project. Every formula here is standard and traceable to the IMF/ILO *Consumer Price Index Manual: Concepts and Methods* (2020) or to a named national statistical institute. If a judge asks “why that formula?”, the answer must be a citation, not a preference.

3.1 Notation

Fixed throughout the document.

Symbol	Meaning
r	Route (ordered city-pair, e.g. DEL→BOM). Directional.
c	Carrier (6E, AI, IX, QP, SG, ...).
τ	Advance-purchase window in days: $\tau \in \mathcal{T} = \{1, 7, 15, 30, 45\}$.
t	Collection date (the day the quote was observed).
d	Departure date. By construction $d = t + \tau$.
s	Source (airline site, OTA, licensed API).
k	Fare family / branded fare (Saver, Flexi, Comfort, ...).
g	Elementary cell (stratum). Default: $g = (r, c, \tau)$.
$p_{i,t}$	Price of quote i observed at time t (INR).
$\tilde{p}_{i,t}$	Cleaned, decomposed price used in the index.
$N_g^{(t)}$	Number of matched quotes in cell g at t .
w_r	Route weight (share of passengers). $\sum_r w_r = 1$.
$\phi_{c r}$	Carrier share <i>within</i> route r . $\sum_c \phi_{c r} = 1$.
ω_τ	Booking lead-time weight. $\sum_\tau \omega_\tau = 1$.
$I_{0,t}^X$	Index of type X comparing period t to base 0, $I_{0,0} = 100$.
$A_g^{(t)}$	Availability ratio in cell g at t (see §4.8).

3.2 Elementary aggregates: the three candidate formulae

At the lowest level no expenditure weights exist, so the index over N matched items in a cell must be an unweighted average of prices or of price relatives. There are exactly three standard choices.

3.2.1 Carli — arithmetic mean of price relatives

$$I_{0,t}^C = \frac{1}{N} \sum_{i=1}^N \frac{p_{i,t}}{p_{i,0}} \quad (3.1)$$

Do not use. Carli fails the *time reversal test*: in general $I_{0,t}^C \cdot I_{t,0}^C > 1$, so it has a known upward bias that compounds when chained. For volatile prices — which airfares emphatically are — the bias is large. Naming Carli and rejecting it for this reason is a cheap way to demonstrate you know the literature.

3.2.2 Dutot — ratio of arithmetic mean prices

$$I_{0,t}^D = \frac{\frac{1}{N} \sum_{i=1}^N p_{i,t}}{\frac{1}{N} \sum_{i=1}^N p_{i,0}} = \frac{\sum_i p_{i,t}}{\sum_i p_{i,0}} \quad (3.2)$$

Dutot passes time reversal, but it *implicitly weights each item by its base price*: an expensive item dominates. Eurostat’s web-scraping guidelines note this explicitly. For airfares this is fatal: a DEL–BOM

quote and a BLR–HYD quote in the same cell would be weighted by their rupee level. Dutot is only admissible *within a homogeneous cell* where all items are genuinely comparable.

3.2.3 Jevons — geometric mean of price relatives

$$I_{0,t}^J = \prod_{i=1}^N \left(\frac{p_{i,t}}{p_{i,0}} \right)^{1/N} = \exp \left(\frac{1}{N} \sum_{i=1}^N \ln \frac{p_{i,t}}{p_{i,0}} \right) \quad (3.3)$$

This is the APIx elementary formula. Justification, in the order to say it out loud:

1. **MoSPI already uses it.** The CPI 2024 FAQ, Q.20: “*The Jevons index (Short index formula) is used for compiling elementary indices in the CPI 2024 series.*” Matching the official formula is the whole integration argument.
2. **It passes time reversal and circularity/transitivity.** $I_{0,t}^J \cdot I_{t,0}^J = 1$ exactly, and $I_{0,1}^J \cdot I_{1,2}^J = I_{0,2}^J$ — so chaining Jevons over a fixed matched sample introduces *no* chain drift.
3. **It is robust to the lognormal shape of fare distributions.** Airfares are strongly right-skewed; taking logs approximately normalises them, and the geometric mean is the natural centre of a lognormal.
4. **It implicitly assumes unitary elasticity of substitution**, which for a consumer choosing between two carriers on the same sector is a far better approximation than Dutot’s zero-substitution or Carli’s inconsistent assumption.

Worked example — build this as a unit test. Cell g has three matched quotes. Base period prices (4000, 6000, 12000); current period (4400, 6000, 9000).

$$I^C = \frac{1}{3}(1.10 + 1.00 + 0.75) = 0.9500 \quad I^D = \frac{4400+6000+9000}{4000+6000+12000} = \frac{19400}{22000} = 0.8818$$

$$I^J = (1.10 \times 1.00 \times 0.75)^{1/3} = (0.8250)^{1/3} = 0.9382$$

Three formulae, the same data, a spread of 6.8 index points. **Put this exact slide in the deck.** It is the fastest possible proof that the index formula is a decision with consequences, not boilerplate. Note also the ordering $I^D < I^J < I^C$, which holds by the AM–GM inequality whenever base prices and relatives are negatively associated.

3.3 Higher-level aggregation: the weighted index family

Above the elementary level, expenditure weights exist and the index becomes a weighted average of elementary indices. Let I_g be the elementary index for cell g and W_g its weight.

3.3.1 Laspeyres — base-period quantity weights

$$I_{0,t}^L = \frac{\sum_g q_{g,0} p_{g,t}}{\sum_g q_{g,0} p_{g,0}} = \sum_g s_{g,0} \frac{p_{g,t}}{p_{g,0}}, \quad s_{g,0} = \frac{p_{g,0} q_{g,0}}{\sum_h p_{h,0} q_{h,0}} \quad (3.4)$$

A Laspeyres index is a base-weighted arithmetic mean of price relatives. It overstates the cost of living because it ignores substitution away from items that became relatively expensive (*substitution bias*).

3.3.2 Paasche — current-period quantity weights

$$I_{0,t}^P = \frac{\sum_g q_{g,t} p_{g,t}}{\sum_g q_{g,t} p_{g,0}} \quad (3.5)$$

Understates, for the mirror-image reason. Requires current-period quantities, which a CPI does not have in real time.

3.3.3 Fisher — the ideal index

$$I_{0,t}^F = \sqrt{I_{0,t}^L \cdot I_{0,t}^P} \quad (3.6)$$

Passes essentially every index-number test (it is “superlative”). Not operationally available for a monthly CPI, but it is the *target* that superlative approximations aim at, and it is the natural benchmark in a back-test.

3.3.4 Törnqvist — the superlative workhorse

$$\ln I_{0,t}^T = \sum_g \frac{s_{g,0} + s_{g,t}}{2} \ln \frac{p_{g,t}}{p_{g,0}} \quad (3.7)$$

A geometric mean of price relatives with average expenditure shares as weights. This is the index that GEKS-Törnqvist (below) makes multilateral, and the one the UK ONS adopted for scanner and alternative data.

3.3.5 Young / Modified Laspeyres — what MoSPI actually uses

A pure Laspeyres needs base-period quantities. In practice statistical offices use weights from an earlier weight-reference period b (here, HCES 2023–24) and prices from a later price-reference period 0 (here, 2024). Two variants:

$$\text{Young: } I_{0,t}^Y = \sum_g s_{g,b} \frac{p_{g,t}}{p_{g,0}} \quad (3.8)$$

$$\text{Lowe (modified Laspeyres): } I_{0,t}^{Lo} = \frac{\sum_g q_{g,b} p_{g,t}}{\sum_g q_{g,b} p_{g,0}} = \sum_g \frac{s_{g,b} \frac{p_{g,0}}{p_{g,b}}}{\sum_h s_{h,b} \frac{p_{h,0}}{p_{h,b}}} \cdot \frac{p_{g,t}}{p_{g,0}} \quad (3.9)$$

The difference is whether the base-period weights are *price-updated* from b to 0. MoSPI CPI 2024 FAQ Q.21: “The Young/Modified Laspeyres’ index is used for compiling higher level indices.” **APIx implements the Young form as default and the Lowe form as a configurable alternative**, because we can price-update our own route weights using our own collected prices — which is a small but real methodological contribution.

The two-line summary a judge will remember: “Jevons at the bottom, Young at the top — exactly what MoSPI does. We didn’t invent a formula; we implemented theirs, and then added the dimensions they don’t have.”

3.4 Chain drift, and why it matters here more than anywhere

A **chained** index multiplies successive period-to-period links:

$$I_{0,T}^{\text{chain}} = \prod_{t=1}^T I_{t-1,t} \quad (3.10)$$

A **direct** (fixed-base) index compares T to 0 in one step. If the formula is transitive, they agree. If not, the difference is **chain drift**.

Drift arises when products enter and leave the sample and prices oscillate — the textbook generator is a promotional cycle where a price falls, sales spike, the price returns, and the index does not. **Airfares are a pathological case:** fares bounce within a day, routes appear and vanish, and the matched sample changes every collection. Diewert’s chapter in the IMF CPI Manual’s companion volume explicitly flags airlines and hotels as the dynamic-pricing case that motivates multilateral methods.

APIx's defence is layered:

1. Jevons on a *strictly matched* sample within a cell is transitive, so no drift arises from the formula itself.
2. Sample churn is handled by **re-matching on a stable product key** (§4.1), not by re-drawing the sample each day.
3. Where churn is unavoidable, we compute a **multilateral** variant and publish the difference as a diagnostic. If the two agree, that is evidence of robustness; if they diverge, that divergence is itself a finding worth reporting.

3.5 Multilateral index methods

Multilateral methods compare *all* periods in a window simultaneously, which makes them transitive by construction and therefore drift-free within the window.

3.5.1 GEKS (Gini–Elteto–Koves–Szulc)

Let $P_{l,m}$ be a bilateral index between periods l and m computed by some formula. Over a window of $T + 1$ periods:

$$P_{0,t}^{\text{GEKS}} = \prod_{l=0}^T \left(P_{0,l} \cdot P_{l,t} \right)^{\frac{1}{T+1}} \quad (3.11)$$

That is: every period is used in turn as a *bridge* between 0 and t , and the geometric mean of all bridged comparisons is taken.

- **GEKS-Jevons (GEKS-J)** uses the Jevons as the bilateral input. This is the correct choice when no expenditure weights exist — Eurostat's HICP web-scraping guidelines name GEKS-Jevons and the Time Product Dummy as the options for weightless web-scraped data. **APIx uses GEKS-Jevons as its multilateral variant.**
- **GEKS-Törnqvist (CCDI)** uses Törnqvist. ONS uses this with a 25-month window and a mean splice for scanner and alternative data. Requires expenditure shares.

3.5.2 Time Product Dummy (TPD)

A regression formulation. Pool all quotes across the window and estimate

$$\ln p_{i,t} = \alpha + \sum_{m=1}^T \delta_m D_m^{\text{time}} + \sum_{j=2}^N \gamma_j D_j^{\text{item}} + \varepsilon_{i,t} \quad (3.12)$$

where D_m^{time} is 1 if the observation is from period m , and D_j^{item} is 1 if the observation is item j . The index is then

$$P_{0,t}^{\text{TPD}} = \exp(\hat{\delta}_t) \quad (3.13)$$

The item dummies absorb permanent quality differences between products; the time dummies carry the pure price movement. With a balanced panel and no weights, TPD and GEKS-Jevons coincide. TPD's real advantage is that it **handles missing observations gracefully** — which for airfares (sold-out cells, cancelled routes) is exactly the situation. A weighted variant (WTPD) uses expenditure shares as regression weights.

Why TPD is a strong demo. It converts index construction into a regression, so you can report R^2 , residual diagnostics, and standard errors on $\hat{\delta}_t$ — i.e. **a confidence interval on the inflation number**. Official CPI does not publish one. Doing so is a visible, defensible piece of added value that costs about forty lines of `statsmodels`.

3.5.3 Geary–Khamis and QU-GK

Geary–Khamis solves simultaneously for a price level P_t and a set of quality-adjusted unit values v_j :

$$v_j = \sum_t \frac{q_{j,t}}{\sum_m q_{j,m}} \cdot \frac{p_{j,t}}{P_t}, \quad P_t = \frac{\sum_j p_{j,t} q_{j,t}}{\sum_j v_j q_{j,t}} \quad (3.14)$$

solved by iteration to a fixed point. ONS’s index-method framework scored **Quality-adjusted Geary–Khamis (QU-GK)** highest when expenditure information is available, and recommended GEKS-Jevons in its absence. **APIx does not have quantities, so GK is documented as future work, not implemented.** Say that plainly; a team that names a method and says “we chose not to implement it, here is why” reads as more competent than one that implements four methods badly.

3.5.4 Splicing: extending a multilateral window

A multilateral index is only transitive *inside* its window. When a new period arrives the window rolls forward, and the new window’s index must be attached to the published series without revising already-published numbers (official CPI is non-revisable). The attachment rule is the *splice*. With P^{new} the freshly-estimated window of length $T + 1$:

Splice	Link factor	Behaviour
Movement (window) splice	$P_{T-1,T}^{\text{new}}$	Uses only the last link. Simple, noisiest.
Window splice	$P_{0,T}^{\text{new}} / P_{0,T-1}^{\text{new}}$	Links on the full window.
Half splice	Link on the window midpoint	Compromise; used by Statbel.
Mean splice	Geometric mean of all possible splice points	Lowest variance; ONS’s choice. APIx default.
FEWS	Fixed-effects index with window splice	Regression-based analogue.

Mean splice, formally, over splice positions $j = 1, \dots, T$:

$$P_{0,T} = P_{0,T-1} \cdot \prod_{j=1}^T \left(\frac{P_{T-j,T}^{\text{new}}}{P_{T-j,T-1}^{\text{new}}} \right)^{1/T} \quad (3.15)$$

3.6 Linking a new series to an old one

MoSPI’s own back-series arithmetic is worth reproducing exactly, because APIx must be able to express its index on the CPI 2024 base and because it is a clean, citable formula.

Let $\bar{I}_y^{\text{new}}$ and $\bar{I}_y^{\text{old}}$ be the annual *geometric mean* indices of the new and old series over a common overlap year y . The linking factor is

$$\text{LF} = \frac{\bar{I}_y^{\text{new}}}{\bar{I}_y^{\text{old}}} \quad \text{and} \quad I_t^{\text{linked}} = I_t^{\text{old}} \times \text{LF} \quad (3.16)$$

For CPI 2024 the overlap year was 2025 and the General-level factors were Rural 0.5222, Urban 0.5320, Combined 0.5267.

Use this in the demo. Show APIx on its own base (APIx Jan-2026 = 100), then apply a linking factor and show it *on the CPI 2024 base* next to the official air fare item index. That single toggle communicates “we understand how your series is actually maintained” better than any amount of dashboard polish.

3.7 The seasonality problem

Airfares are violently seasonal: Diwali, Christmas–New Year, summer holidays, wedding season, Kumbh-scale events. Three treatments, and APIx publishes all three:

1. **Not seasonally adjusted (NSA).** The raw index. This is what CPI publishes (“2024=100, NSA”).
2. **Year-on-year rate.** $\pi_t = (I_t/I_{t-12} - 1) \times 100$ — differencing at lag 12 removes stable seasonality without a model. This is exactly MoSPI’s published inflation definition:

$$\text{Inflation}(\%) = \frac{I_t - I_{t-12}}{I_{t-12}} \times 100 \quad (3.17)$$

3. **Model-based SA.** X-13ARIMA-SEATS or STL decomposition, with an explicit *moving-holiday regressor* for Diwali (which shifts across the Gregorian calendar) and event dummies for known demand shocks. Report SA as *experimental* — a series with under two years of history cannot support a credible seasonal factor, and saying so is the statistically literate answer.

3.8 Sampling and variance

3.8.1 The estimator and its uncertainty

A published index is a point estimate from a sample. Its sampling variance is estimable, and publishing it is a differentiator. For the Jevons, work in logs where the estimator is a simple mean:

$$\hat{\mu}_g = \frac{1}{N_g} \sum_i \ln \frac{p_{i,t}}{p_{i,0}}, \quad \widehat{\text{Var}}(\hat{\mu}_g) = \frac{1}{N_g(N_g - 1)} \sum_i \left(\ln \frac{p_{i,t}}{p_{i,0}} - \hat{\mu}_g \right)^2 \quad (3.18)$$

and by the delta method the variance on the index level $I_g^J = \exp(\hat{\mu}_g)$ is

$$\widehat{\text{Var}}(I_g^J) \approx (I_g^J)^2 \widehat{\text{Var}}(\hat{\mu}_g) \quad (3.19)$$

Aggregating with weights W_g (treating cells as independent strata):

$$\widehat{\text{Var}}(I) = \sum_g W_g^2 \widehat{\text{Var}}(I_g) \quad (3.20)$$

3.8.2 Bootstrap intervals

Because fare distributions are neither normal nor independent within a flight, the analytic interval is indicative only. APIx also ships a **block bootstrap**: resample *whole flights* (not individual quotes) with replacement, $B = 1000$ times, recompute the full index pipeline, and take the 2.5th and 97.5th percentiles. Blocking at the flight level preserves the within-flight correlation induced by shared inventory.

3.8.3 Sample size for a target precision

To detect a $\delta\%$ move at significance α with power $1 - \beta$, in log space with per-quote log-relative standard deviation σ :

$$N \geq \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2 \sigma^2}{(\ln(1 + \delta/100))^2} \quad (3.21)$$

Estimate σ from the first week of live collection and **publish the resulting minimum cell size as a design constraint** — cells that fall below it are suppressed rather than published noisily. Suppression rules are what official statistics does; mimicking them signals maturity.

PART 4

The APIx Specification

This chapter is the formal definition of the index. It should be readable as a specification document by someone who has never seen the code, and every step should be implementable and testable in isolation. Treat it as the contract between the statistics work and the engineering work.

4.1 The product key: what exactly is being priced

A price index requires a *constant-quality* product whose price is tracked over time. For airfares this definition is the whole game — get it wrong and every downstream number is meaningless. APIx defines the priced product as:

One economy-class, one-way, one-adult, hand-baggage-inclusive seat on a non-stop scheduled flight, on route r , operated by carrier c , departing on date d , quoted at exactly τ days before departure, at a fixed collection clock time, from an unauthenticated India-egress session.

Each qualifier is load-bearing:

Qualifier	Why it is in the definition
Economy class	GST differs by cabin (5% economy vs. 12% other than economy — [VERIFY]); mixing cabins mixes tax regimes and quality.
One-way	Return fares are bundled products with their own pricing logic; a separate return-basis index is a documented extension.
One adult	Child/infant fares follow different rules; multi-pax triggers group pricing.
Hand baggage only	The minimum common bundle across LCC and FSC. Checked bag is an ancillary and is indexed separately.
Non-stop	A one-stop itinerary is a different service (longer duration = lower quality). Mixing them is unadjusted quality change.
Fixed τ	The advance-purchase window is a product characteristic. A T+7 fare and a T+45 fare are different products, not two observations of one.
Fixed clock time	Fares move intraday. Collecting at 02:00 IST one day and 14:00 the next injects diurnal noise into the price signal.
Unauthenticated, cookie-clean	Removes personalisation and loyalty pricing. A logged-out session is also the legally cleanest one (Part IV).
India egress	Currency, tax treatment and price display are geography-dependent.

4.1.1 The matching key

Two quotes are *matched* (i.e. the same product in two periods) if and only if they agree on the tuple

$$\kappa = (r, c, \tau, \text{flight_number}, \text{fare_family } k, \text{cabin}, \text{stops}) \quad (4.1)$$

Matching on `flight_number` rather than on departure date is deliberate: it tracks *the same flight in the schedule* (e.g. 6E-2134, the 07:10 DEL→BOM) across weeks, which is the closest airline analogue of tracking the same SKU in a shop. It also survives the daily rotation of departure dates that τ being fixed necessarily implies.

Day-of-week is a confound and must be controlled. If τ is fixed and t advances one day, then d advances one day too — so a Tuesday-departure observation becomes a Wednesday-departure observation. Since fares are strongly day-of-week dependent, a naive daily index would report

pure day-of-week variation as inflation.

APIx handles this in two ways: (i) the daily index is computed on a **7-day centred geometric moving average** of the daily cell relatives, which annihilates a weekly cycle exactly; and (ii) a day-of-week fixed effect is included in the TPD specification. State this unprompted — it is the kind of trap a MoSPI statistician will probe for, and having the answer ready is worth more than any feature.

4.2 Stage 0: raw quote to cleaned price

4.2.1 Fare decomposition

A displayed “total fare” on an Indian OTA is a sum of economically distinct components with completely different inflation dynamics. Indexing the total without decomposing it means a UDF revision at one airport shows up as “airfare inflation”. The decomposition:

$$P^{\text{total}} = \underbrace{P^{\text{base}}}_{\text{airline, dynamic}} + \underbrace{\text{YQ} + \text{YR}}_{\text{carrier surcharges}} + \underbrace{\text{UDF} + \text{ASF} + \text{RCS}}_{\text{statutory / airport}} + \underbrace{\text{GST}}_{\text{tax}} + \underbrace{\text{CF}}_{\text{OTA convenience fee}} \quad (4.2)$$

Component	Set by	Treatment in APIx
P^{base}	Airline revenue management	The core signal. Primary index is built on this.
YQ / YR	Airline (fuel / carrier misc.)	Economically part of the fare; folded into P^{base} for the headline index, reported separately in the diagnostic view.
UDF	AERA tariff orders, per airport	Administered price. Changes are step functions on known dates. Indexed separately.
ASF	BCAS / MoCA notification	Administered, uniform. Same treatment.
RCS levy	MoCA (UDAN cross-subsidy)	Applies on specified routes. Same treatment.
GST	GST Council; 5% economy / 12% other [VERIFY]	Ad valorem on the fare base.
CF	The OTA	Not a fare. It is a distribution charge that differs across OTAs for an identical seat. Excluded from the fare index; published as a separate “Distribution Cost” series.

APIx publishes **three headline variants**, which is itself a USP because it lets a user see exactly how much of “airfare inflation” is airline behaviour versus administered charges:

$$\text{APIx-B : base fare} + \text{YQ/YR only} \quad (\text{pure airline pricing}) \quad (4.3)$$

$$\text{APIx-T : everything except OTA convenience fee} \quad (\text{what a traveller pays — headline}) \quad (4.4)$$

$$\text{APIx-A : APIx-T} + \text{OTA convenience fee} \quad (\text{all-in checkout cost}) \quad (4.5)$$

Do not scrape administered charges. UDF, ASF, RCS and GST rates are published by AERA, BCAS/MoCA and the GST Council. Maintain them as a **slowly-changing dimension table** loaded from official notifications, versioned with effective-from/effective-to dates. Then compute the base fare as a residual and cross-check it against the airline’s own fare breakup where one is displayed. This turns a scraping problem into a reconciliation problem — and reconciliation is auditable.

4.2.2 Outlier detection

Fare quotes contain genuine extremes (last-seat pricing) and genuine errors (parse failures, currency confusion, promotional glitches). The cleaning rule must remove the second without deleting the first, because the first is the signal.

Work in log space throughout: $y_i = \ln \tilde{p}_i$.

1. **Hard validity gates** (deterministic, not statistical). Reject if: $\tilde{p} \leq 0$; $\tilde{p} < \text{the sum of known statutory components}$; currency $\neq$ INR; τ mismatch > 0 days; duration implausible for the great-circle distance; flight number not in the schedule reference.
2. **Tukey fences within cell**, on logs. With Q_1, Q_3 the cell quartiles and $\text{IQR} = Q_3 - Q_1$:

$$\text{flag if } y_i < Q_1 - 3 \text{ IQR} \quad \text{or} \quad y_i > Q_3 + 3 \text{ IQR} \quad (4.6)$$

The multiplier is 3, not the usual 1.5: fares are legitimately dispersed and 1.5 would trim real observations.

3. **Hampel / MAD filter** on the *price relative*, which is more robust than quartiles for small cells. With m the median and $\text{MAD} = \text{median}_i |y_i - m|$:

$$z_i^{\text{rob}} = \frac{0.6745 (y_i - m)}{\text{MAD}}, \quad \text{flag if } |z_i^{\text{rob}}| > 3.5 \quad (4.7)$$

(The constant 0.6745 makes MAD a consistent estimator of σ under normality.)

4. **Longitudinal jump test**. Flag a quote whose own log-relative to its previous observation exceeds a cell-specific threshold: $|\ln(p_{i,t}/p_{i,t-1})| > 3\hat{\sigma}_g$. This catches parse regressions that survive the cross-sectional filters.
5. **Disposition, not deletion**. A flagged quote is **never silently dropped**. It is written to the store with `quality_flag` set and an explicit disposition in {ACCEPTED, WINSORISED, EXCLUDED, QUARANTINED}. The published index uses `ACCEPTED` $\cup$ `WINSORISED`; the audit endpoint exposes all four. Winsorisation caps at the fence rather than removing:

$$y_i^w = \min(\max(y_i, Q_1 - 3 \text{ IQR}), Q_3 + 3 \text{ IQR}) \quad (4.8)$$

4.2.3 Imputation for missing quotes

If a cell has a matched product that yields no quote at t (page failed, flight cancelled, inventory sold out), leaving a gap biases the index. Three cases, three treatments:

Cause	Detectable by	Treatment
Collection failure	HTTP error, empty parse, circuit-breaker trip	Retry, then impute by cell-relative carry-forward: $\hat{p}_{i,t} = p_{i,t-1} \times I_g^{t-1,t}$, i.e. move the missing item by its own cell's observed change. Flag as imputed.
Flight not operating (schedule change, cancellation)	Absent from schedule reference	Remove from the matched sample prospectively. A flight that does not exist has no price. Do not impute.
Sold out / fare class closed	Present in schedule, absent from availability	This is a price signal, not missing data. See §4.8.

4.3 Stage 1: elementary index

For each cell $g = (r, c, \tau)$ with matched quote set $\mathcal{M}_g$ present in both 0 and t :

$$I_{g,t} = 100 \times \exp \left(\frac{1}{|\mathcal{M}_g|} \sum_{i \in \mathcal{M}_g} \ln \frac{\tilde{p}_{i,t}}{\tilde{p}_{i,0}} \right) \quad (4.9)$$

Suppress and mark the cell if $|\mathcal{M}_g| < N_{\min}$ (default $N_{\min} = 5$, derived from the power calculation of §3.8.3). A suppressed cell's weight is redistributed pro-rata to the remaining cells within the same route — exactly the mechanism MoSPI uses when an item is unavailable in a market.

4.4 Stage 2: carrier aggregation within a route

$$I_{r,\tau,t} = \sum_c \phi_{c|r} I_{(r,c,\tau),t} \quad (4.10)$$

with the within-route carrier weight

$$\phi_{c|r} = \frac{S_c \cdot F_{c,r} \cdot \text{Seats}_{c,r} \cdot \text{PLF}_c}{\sum_{c'} S_{c'} \cdot F_{c',r} \cdot \text{Seats}_{c',r} \cdot \text{PLF}_{c'}} \quad (4.11)$$

where $F_{c,r}$ is weekly frequency on the route, $\text{Seats}_{c,r}$ the average seats per departure, PLF_c the carrier's published passenger load factor, and S_c an optional national-share prior for smoothing thin routes. In practice $F \times \text{Seats} \times \text{PLF}$ is an estimate of *passengers actually carried by carrier c on route r* , which is the economically correct weight.

Why this beats equal weighting. IndiGo carries roughly two thirds of Indian domestic passengers (67.4% in July 2026). Under equal carrier weighting, a fare move at SpiceJet (1.6% share) would move the index as much as one at IndiGo. Under $\phi_{c|r}$, it moves it proportionally to the passengers actually exposed to it. Load factors are published by DGCA monthly (Akasa 92.2%, SpiceJet 87.8%, AI Group 85.5%, IndiGo 85.1% in June 2026), so this weight is fully sourced.

4.5 Stage 3: route aggregation

$$I_{\tau,t} = \sum_{r \in \mathcal{B}} w_r I_{r,\tau,t}, \quad w_r = \frac{Q_r}{\sum_{r' \in \mathcal{B}} Q_{r'}} \quad (4.12)$$

where $\mathcal{B}$ is the route basket and Q_r is annual passengers on route r from DGCA/AAI sector-level traffic. Where the PSD supplies its own route list and weights — which is precisely what the PS means by “*an index-construction module based on PSD given routes and weights*” — the module accepts them as a signed input file and uses them in place of the derived weights. **Build the weight vector as an injectable parameter from day one;** the whole point is that PSD, not the team, owns the weights.

4.6 Stage 4: lead-time aggregation and the headline

$$\text{APIx}_t = \sum_{\tau \in \mathcal{T}} \omega_\tau I_{\tau,t} \quad (4.13)$$

ω_τ **is the honest weak point of this project, and you should say so.** The weights over advance-purchase windows should be the *distribution of actual booking lead times of Indian domestic travellers*. That distribution is held by the airlines and the OTAs and is not published. Note that the DGCA asked airlines in December 2024 for exactly this kind of ticket-level data — booking date, fare charged, base fare — and the Federation of Indian Airlines refused on commercial-confidentiality grounds. So the data exists and is contested; it is not something a hackathon team can obtain.

APIx's answer: treat ω_τ as a declared, versioned policy parameter, ship three named presets (*Uniform, Leisure-tilted, Business-tilted*), publish the index under all three, and include a tornado sensitivity chart showing how much the headline moves across the range. Then state that MoSPI or DGCA can replace it with a measured distribution the moment one exists. Turning your biggest gap into a published sensitivity analysis is a strength move, not a weakness.

4.7 The dual time basis

This is APIx’s most distinctive methodological contribution, and it is directly grounded in international practice.

Basis	Definition and rationale
Booking basis (acquisition approach)	The quote is attributed to the <i>collection date</i> t . Answers: “ <i>what does it cost to buy a flight today?</i> ” This is the consumer-facing, regulator-facing, RBI-nowcast-facing series.
Travel basis (use approach)	The quote is attributed to the <i>departure date</i> $d = t + \tau$. Answers: “ <i>what is the price of air travel consumed in this month?</i> ” This is the CPI-consistent series. The UK ONS states explicitly that changes in air fares are recorded in the CPI in the month in which the flight departs, not when the ticket is bought .

Formally, the travel-basis index at month m aggregates over all quotes whose departure falls in m , weighted by the lead-time distribution:

$$\text{APIx}_m^{\text{travel}} = \sum_{\tau \in \mathcal{T}} \omega_{\tau} I_{\tau}(t = m - \tau) \quad (4.14)$$

whereas the booking-basis index is

$$\text{APIx}_m^{\text{book}} = \sum_{\tau \in \mathcal{T}} \omega_{\tau} I_{\tau}(t = m) \quad (4.15)$$

The two differ by a *lead-time-weighted lag structure*. Publishing both and showing the reconciliation is a genuinely novel contribution for an Indian price series, and it directly demonstrates that the team read foreign NSI methodology rather than reinventing one badly.

4.8 Availability adjustment

4.8.1 The bias

Suppose a cell normally offers fare families at Rs. 4,000, Rs. 6,000 and Rs. 9,000. During a demand surge the Rs. 4,000 and Rs. 6,000 buckets sell out. A naive matched index observes only the Rs. 9,000 quote — but that quote has not *changed price*, so a strictly matched index reports **zero inflation** during the exact episode a consumer experiences as a fare explosion. This is not a hypothetical; it is the mechanism by which any matched-model fare index understates surge periods, and it is the single most defensible methodological criticism of a naive implementation.

4.8.2 The APIx treatment

Define the availability ratio for cell g at t :

$$A_g^{(t)} = \frac{\# \text{ fare families observed offered at } t}{\# \text{ fare families in the cell's reference bundle}} \in (0, 1] \quad (4.16)$$

Then compute a **lowest-available-fare (LAF) index** alongside the matched index. The LAF index tracks $\min_k \tilde{p}_{i,t,k}$ within the flight, i.e. the cheapest seat a consumer could actually buy:

$$I_{g,t}^{\text{LAF}} = 100 \times \exp \left(\frac{1}{|\mathcal{M}_g|} \sum_i \ln \frac{\min_k \tilde{p}_{i,t,k}}{\min_k \tilde{p}_{i,0,k}} \right) \quad (4.17)$$

The **Availability-Adjusted APIx** is a geometric blend controlled by A :

$$I_{g,t}^{\text{adj}} = (I_{g,t}^{\text{matched}})^{A_g^{(t)}} \cdot (I_{g,t}^{\text{LAF}})^{1-A_g^{(t)}} \quad (4.18)$$

When everything is available ($A = 1$) the adjusted index equals the matched index and nothing changes. As buckets close ($A \rightarrow 0$) it converges to the lowest-available-fare index, which does capture the surge. The blend is continuous, monotone, and reduces exactly to standard practice in the normal case — which is what makes it defensible rather than ad hoc.

This is the single best demo in the whole project. Pick a real surge — a festival weekend, or a disruption episode — and show three lines on one chart: the naive matched index (flat, wrong), the LAF index (spiking), and the availability-adjusted index (spiking, correctly). Then show the $A_g^{(t)}$ series underneath as an area chart collapsing toward zero. Ninety seconds, and it proves the team understands price measurement rather than data plumbing.

4.9 The lead-time elasticity surface

4.9.1 The model

Estimate the fare surface by pooled log-linear regression with fixed effects:

$$\ln \tilde{p}_i = \alpha + \beta_1 \ln \tau_i + \beta_2 (\ln \tau_i)^2 + \gamma_r + \lambda_c + \theta_{\text{dow}(d_i)} + \psi_{\text{month}(d_i)} + \rho \text{Holiday}_i + \varepsilon_i \quad (4.19)$$

where γ_r , λ_c , θ , ψ are route, carrier, day-of-week and month-of-departure fixed effects and Holiday_i is a festival/long-weekend indicator. Standard errors clustered by flight.

4.9.2 The elasticity

The *lead-time elasticity of fare* is the derivative of log price with respect to log lead time:

$$\eta(\tau) = \frac{\partial \ln p}{\partial \ln \tau} = \beta_1 + 2\beta_2 \ln \tau \quad (4.20)$$

Interpretation: $\eta(\tau) = -0.30$ at $\tau = 7$ means a 10% increase in booking lead time around one week out is associated with a 3% lower fare. Because β_2 enters, the elasticity is allowed to *vary with lead time* — which matters, since fares typically fall from $\tau = 45$ to a trough somewhere around $\tau = 21$ –30 and then rise sharply approaching departure. The trough is at

$$\tau^* = \exp\left(-\frac{\beta_1}{2\beta_2}\right) \quad (4.21)$$

which yields a directly publishable consumer statistic: **“the cheapest time to book DEL–BOM is τ^* days out.”** That single number is the most press-friendly output of the entire system.

4.9.3 Fare-bucket detection

Because airlines price from discrete RBDs, the empirical fare distribution within a route–carrier is multi-modal. Fit a one-dimensional Gaussian mixture in log space:

$$f(y) = \sum_{j=1}^J \pi_j \mathcal{N}(y; \mu_j, \sigma_j^2), \quad \sum_j \pi_j = 1 \quad (4.22)$$

with J chosen by BIC. The fitted $\{\mu_j\}$ are estimates of the **fare ladder** — the airline’s actual bucket structure, recovered from public prices without any privileged data. Tracking μ_j over time separates “the airline raised its fare ladder” (structural) from “more people bought so we are selling from a higher bucket” (compositional). Very few teams will attempt this and it is directly relevant to DGCA’s tariff-compliance mandate under Rule 135 of the Aircraft Rules, 1937, which requires airlines to charge within their declared tariff bands.

4.10 Nowcasting CPI Transport

APIx is available daily; the CPI air fare item index is monthly, published around the 12th of the following month. That gap is a forecasting opportunity.

Specify a mixed-frequency bridge. Let π_m^{CPI} be the monthly y-o-y inflation of the air fare item and x_m the monthly average of the APIx travel-basis daily series:

$$\pi_m^{\text{CPI}} = \alpha + \sum_{j=0}^J \beta_j x_{m-j} + \sum_{k=1}^K \rho_k \pi_{m-k}^{\text{CPI}} + \delta \text{ATF}_m + u_m \quad (4.23)$$

with an ATF-price control (ATF is 35–40% of airline operating cost, and PPAC publishes prices). Report:

- Out-of-sample RMSE against a naive random-walk benchmark. If APIx does not beat a random walk, say so — a negative result honestly reported is more credible than an unvalidated claim.
- A Granger-causality test of APIx $\rightarrow$ CPI air fare.
- Lead-lag cross-correlation at daily and monthly frequency.

Do not claim a nowcast of *headline* CPI. Air fare is a small slice of a division that is 8.796% of combined CPI. Claim a nowcast of the **air fare item index** and, at most, a contribution to the Transport division. Overclaiming here is the fastest way to lose a statistician judge.

4.11 The full computation, end to end

Step	Operation	Formula / rule
0a	Validity gates	Deterministic rejects
0b	Decompose	$P^{\text{base}} = P^{\text{total}} - \text{levies} - \text{GST} - \text{CF}$
0c	Outlier flag	Tukey $3 \times \text{IQR}$ and Hampel $ z^{\text{rob}} > 3.5$ on logs
0d	Impute / winsorise	Cell-relative carry-forward; cap at fences
1	Elementary index	Jevons over matched κ within $g = (r, c, \tau)$
1b	Availability adjust	$I^{\text{adj}} = (I^{\text{matched}})^A (I^{\text{LAF}})^{1-A}$
1c	De-seasonalise DOW	7-day centred geometric moving average
2	Carrier aggregation	$\phi_{c r}$ from frequency $\times$ seats $\times$ PLF
3	Route aggregation	w_r from DGCA sector traffic, or PSD-supplied
4	Lead-time aggregation	ω_τ , three declared presets
5	Rebase and link	$\times 100$; linking factor to CPI 2024 base
6	Publish	Daily, weekly (7-day GM), monthly (calendar-month GM)
7	Uncertainty	Delta-method SE + flight-block bootstrap CI
8	Robustness	GEKS-Jevons and TPD variants; publish the spread

Monthly aggregation uses the *geometric* mean of daily indices, consistent with MoSPI's own use of geometric means for annual averages in the linking-factor computation:

$$\text{APIx}_m = \left(\prod_{t \in m} \text{APIx}_t \right)^{1/|m|} \quad (4.24)$$

PART 5

Data: Sources, Basket, and Schema

5.1 The source inventory

Every source below is either an official Government of India publication, a licensed commercial API with a free tier, or an open dataset. **Verify every URL the week of the build** — government portals reorganise.

5.1.1 Official statistics and administrative data

Source	What it gives	Where
MoSPI e-Sankhyiki	CPI 2024 indices: General, Division, Group, Class, Sub-class, item; All-India and State/UT; monthly	https://esankhyiki.mospi.gov.in/macroeconomicindicators?product=cpi
MoSPI CPI portal	Item-wise weights, methodology notes, announcements	https://cpi.mospi.gov.in
MoSPI press releases	Monthly CPI PDFs with annexures, response rates, linking factors	https://www.mospi.gov.in
Expert Group Report on Comprehensive Updation of CPI	The authoritative methodology document for CPI 2024: COICOP mapping, online markets, alternative data sources	Linked from the CPI 2024 FAQ; https://cpi.mospi.gov.in
HCES 2023–24	Household consumption expenditure, MMRP; the weight source	MoSPI microdata catalogue
DGCA monthly traffic report	Passengers carried by carrier, market share, PLF, cancellations, complaints, OTP	https://www.dgca.gov.in → Statistics / Monthly Traffic
AAI traffic statistics	Airport-wise and sector-wise passenger movements — the basis for route weights w_r	https://www.aai.aero → Traffic News
AERA tariff orders	UDF per airport, effective dates	https://aera.gov.in
MoCA / BCAS	ASF notifications, RCS-UDAN levy, Rule 135 tariff filing framework	https://www.civilaviation.gov.in
PPAC	ATF and fuel retail selling prices — the cost-side driver and a nowcast regressor	https://ppac.gov.in
Parliament Q&A	Ministerial replies containing TMU coverage, fare-movement statistics	https://sansad.in , MoCA site
RBI DBIE	CPI series, policy rates, for the macro overlay	https://data.rbi.org.in

5.1.2 Fare data: the source ladder, from most to least defensible

Rung	Channel	Notes	Legal basis
1	Licensed distribution APIs — Amadeus Self-Service, Duffel, Kiwi Tequila, Travelpayouts, Skyscanner partner API, OAG	Free/sandbox tiers exist and are sufficient for a prototype. Return structured fare, carrier, cabin, fare family and often a tax breakdown. <i>This should be the primary source.</i>	Contract; explicit permission
2	Airline NDC / partner feeds	Where obtainable. Best fidelity on fare families.	Contract
3	Public pages that permit automated access , verified against <code>robots.txt</code> at collection time, with declared User-Agent and low rate	Only for hosts whose <code>robots.txt</code> does not disallow the relevant path and whose terms do not prohibit it. Logged-out only.	Implied licence; <code>robots.txt</code>
4	Aggregated third-party datasets — OAG, Cirium, ARC, Sabre market intelligence	Commercial. Named in the roadmap as the production path; not affordable in a hackathon.	Contract
X	Anything requiring circumvention — CAPTCHA solving, residential-proxy rotation to defeat blocking, fake accounts, header spoofing to impersonate a browser after a block	Explicitly out of scope. See §6.2.	None

5.1.3 Reference and enrichment data

Dataset	Use
OurAirports / OpenFlights	IATA and ICAO codes, coordinates, timezone; great-circle distance for the plausibility gate
Aircraft type → seat count	Estimating $\text{Seats}_{c,r}$ for the carrier weight
Indian public holiday & festival calendar	The Holiday_i regressor; moving-holiday adjustment for Diwali
School vacation calendars (CBSE/state)	Leisure-demand seasonality
Great-circle distance	Fare-per-km normalisation; enables the “India’s fare per km” comparison the Minister used in Parliament
GST rate schedule	Ad valorem component; SCD table

5.2 Designing the route basket

5.2.1 Selection rule

The PS names an illustrative basket (DEL–BOM, DEL–BLR, BOM–BLR, DEL–CCU, BLR–HYD, MAA–DEL). Do not just hard-code it. Derive it, then show that the derivation reproduces the PS list — which simultaneously satisfies the PS and demonstrates method.

1. Rank all directional city-pairs by annual passengers Q_r from AAI/DGCA sector data.
2. Select the top M routes such that cumulative coverage exceeds a target:

$$\mathcal{B} = \arg \min_M \left\{ M : \frac{\sum_{r=1}^M Q_{(r)}}{\sum_{\text{all}} Q_r} \geq \theta \right\} \quad (5.1)$$

with $\theta = 0.60$ for the hackathon build and $\theta = 0.80$ documented as the production target. Report

the achieved coverage as a headline metric — and note that the DGCA TMU’s 78 routes cover about 27%, so beating that number is a concrete, quotable win.

3. **Stratify, do not just take the head.** Force representation across: metro–metro, metro–tier-2, tier-2–tier-2, and **UDAN/RCS routes** (which are subsidised and price-capped, and therefore behave completely differently — including them is a genuine methodological point about heterogeneity).
4. Force geographic spread: North, West, South, East, North-East. A basket that is all golden-quadrilateral routes is not a national index.
5. Re-select annually; publish the basket and the selection date. Mid-series basket changes must be linked, not spliced silently.

5.2.2 The build-day basket

Stratum	Example sectors	Target count	Why in the basket
Metro trunk	DEL–BOM, DEL–BLR, BOM–BLR, DEL–CCU, DEL–MAA, BOM–DEL	10–12	Highest Q_r ; the PS’s named sectors
Metro–tier 2	DEL–LKO, BOM–AMD, BLR–COK, CCU–BBI	8–10	Where the marginal Indian flyer actually is
Tier2–tier2	AMD–PNQ, JAI–LKO	3–4	Thin routes, different competitive structure
North-East	DEL–GAU, CCU–IXA	2–3	Route Dispersal Guidelines category; policy salience
UDAN / RCS	Select capped sectors	2–3	Administered pricing — a natural control group
Leisure/seasonal	DEL–GOI, DEL–SXR, DEL–IXL	3–4	Extreme seasonality; stress-tests the SA module

Both directions of each pair are separate routes. $M \approx 30$ directional routes $\times$ 5 carriers $\times$ 5 windows = 750 cells; at ≈ 6 flights per cell that is roughly **4,500 quotes per collection cycle** — comfortably inside a polite request budget when sourced primarily from APIs.

5.3 Data model

5.3.1 Star schema

Table	Type	Key columns
fact_fare_quote	Fact (hypertable)	quote_id (uuid pk), collected_at_utc, collected_at_ist, route_sk, carrier_sk, source_sk, flight_number, departure_date, departure_time_local, arrival_time_local, apw_days, cabin, fare_family, rbd, stops, currency, total_fare, base_fare, yq_yr, udf, asf, rcs_levy, gst, convenience_fee, seats_remaining_shown, is_sold_out, quality_flag, disposition, provenance_id, raw_hash
dim_route	Dimension	route_sk, origin_iata, dest_iata, origin_city, dest_city, gc_distance_km, stratum, is_rcs, annual_pax, weight_wr, basket_version
dim_carrier	Dimension	carrier_sk, iata, icao, name, model (LCC/FSC), national_share, plf, as_of_month
dim_source	Dimension	source_sk, name, type (api/site), ladder_rung, legal_basis, tos_url, robots_snapshot_id
dim_date	Dimension	date_sk, dow, is_holiday, festival_name, is_long_weekend, school_vacation_flag
scd_airport_charges	SCD Type 2	airport_iata, udf_domestic, asf, effective_from, effective_to, source_notification_url
scd_tax_rate	SCD Type 2	cabin, gst_rate, effective_from, effective_to, notification_ref
fact_index_value	Fact	index_id, index_code (APIx-B/T/A, by r, τ), period, frequency, value, se, ci_low, ci_high, n_quotes, n_cells, coverage_pct, method_version, weights_version, run_id
ledger_provenance	Append-only	provenance_id, run_id, source_sk, request_url_hash, robots_directive_applied, rate_limit_bucket, http_status, fetched_at, prev_hash, this_hash

5.3.2 Design decisions worth defending

- **Quotes are immutable.** No `UPDATE` on `fact_fare_quote`, ever. Corrections are new rows with a supersession pointer. Official statistics is an append-only discipline; a mutable price table is disqualifying.
- **quality_flag and disposition are separate.** The flag says what the detector found; the disposition says what the pipeline did about it. Conflating them loses the ability to re-run cleaning with different thresholds.
- **raw_hash** is the SHA-256 of the canonicalised raw payload, stored so a number can be traced to the exact bytes it came from without retaining the full payload indefinitely.
- **method_version and weights_version** are stamped on every index value. When the methodology changes, old numbers remain reproducible under the old version.
- **TimescaleDB hypertable** on `collected_at_utc` with 1-day chunks, compression after 7 days, continuous aggregates for the daily and weekly rollups.
- **Both UTC and IST** are stored. Fares are quoted against local departure times; collection scheduling is in IST; analysis needs monotone UTC. Storing one and deriving the other has caused more price-statistics bugs than any other single decision.

5.4 Data quality contract

Implement as executable assertions (Great Expectations, or plain `pytest` against a `pandera` schema). The suite runs on every ingestion batch and blocks publication on failure.

Check class	Assertion	On failure
Schema	Types, nullability, enum domains for cabin/disposition	Block
Range	$0 < \text{base fare} < \text{total fare}$; $\text{apw_days} \in \mathcal{T}$; $\text{GST} \approx \text{rate} \times \text{base}$	Block
Arithmetic	$ P^{\text{total}} - \sum \text{components} \leq \text{INR } 1$	Block
Freshness	Every basket route has ≥ 1 quote in the last 26 h	Alert
Completeness	Cell fill rate $\geq 80\%$ of expected	Alert; suppress cell
Uniqueness	No duplicate $(\kappa, t, \text{source})$	Block
Distributional	Cell median log-fare within 4σ of trailing 28-day median	Quarantine
Referential	Every <code>provenance_id</code> resolves; hash chain intact	Block
Cross-source	Same flight from two sources differs by $\leq 5\%$ ex-convenience-fee	Alert

The cross-source check is a hidden USP. When two independent sources quote the same flight, the difference is an estimate of *measurement error* in your own instrument. Publishing a measurement-error series — “our quotes agree to within $x\%$ ” — is something no competing team will think of, and it is precisely the sort of thing a statistical office cares about.

PART 6

System Architecture and Engineering

6.1 Architecture at a glance

SOURCES	COLLECTION	STORAGE	COMPUTE	SERVING
+-----+ Licensed fare APIs -(1)> +-----+	+-----+ SOURCE LADDER resolver +-----+ POLITENESS GOV. - robots.txt OTA pages -(3)> (permitted) +-----+ - UA declared --> +-----+ - backoff +-----+ - circuit brkr +-----+ AERA/BCAS PPAC/DGCA -(0)> AAI/MoSPI +-----+	+-----+ Postgres16 +Timescale +-----+ fact_fare_ quote +-----+ dim_* +-----+ scd_* +-----+ ledger_ +-----+ provenance +-----+ Parquet lake (DuckDB) +-----+ v +-----+ Great Expect. quality gates -----> +-----+	+-----+ Cleaning validity +-----+ outliers imputing +-----+ v +-----+ INDEX ENGINE +-----+ Jevons Young/Lowe +-----+ GEKS-J, TPD Avail.adj. +-----+ Bootstrap +-----+ v +-----+ fact_index_ value +-----+	+-----+ FastAPI REST + SDMX +-----+ /index /routes +-----+ /elasticity /components +-----+ /coverage /provenance +-----+ v +-----+ React SPA Recharts heat map surface +-----+
ORCHESTRATION Celery beat/APScheduler OBSERVABILITY Prometheus+Grafana+Sentry				
PACKAGING Docker Compose / K8s manifests CI GitHub Actions				

6.2 The ethical collection framework

6.2.1 The legal position, accurately stated

India has **no statute that either permits or prohibits web scraping**. Liability is assembled from four instruments:

Instrument	Exposure
IT Act, 2000, s.43	Compensation for a person who, <i>without permission of the owner</i> , accesses a computer resource or downloads, copies or extracts data from it. Also covers disruption — so excessive request volume is independently actionable.
IT Act, 2000, s.66	Turns the s.43 conduct into a criminal offence where done dishonestly or fraudulently.
Copyright Act, 1957	Facts are not protected, but a <i>compilation</i> can be, if it satisfies the “skill and judgment” standard — the test the Supreme Court applied in <i>Eastern Book Company v. D.B. Modak</i> . Fare tables plausibly qualify.
Indian Contract Act, 1872	Terms of use are an enforceable contract. Breaching a no-automated-access clause is a contract claim independent of the IT Act.
DPDP Act, 2023	Engages only if personal data is collected. APIx collects no personal data — fares are not personal data — which removes this exposure entirely, and saying so explicitly is a strong point.

Two international precedents worth knowing because judges may raise them: *hiQ Labs v. LinkedIn* (9th Cir.) held that scraping public pages is not “unauthorised access” under the US CFAA; *Meta v. Bright Data* (2024) declined to find breach for scraping *logged-out* public pages, on the reasoning that terms bind account holders. The emergent rule — **logged-out public collection is defensible; logged-in collection against accepted terms is not** — is directly encoded in the APIx product key, which mandates unauthenticated sessions.

6.2.2 The Zero-Circumvention Policy

APIx will not, under any configuration: solve, bypass or outsource a CAPTCHA; rotate residential or datacentre IPs for the purpose of evading a block; create or use accounts to obtain fares; spoof headers or fingerprints to impersonate a human after being blocked; or continue requesting a host after a 429, 403 or explicit block.

APIx will: read and cache `robots.txt` per host and honour it as binding policy; declare a truthful, contactable User-Agent (`APIx-Research/1.0 (+contact-url)`); enforce a per-host token bucket well below any published crawl-delay; exponentially back off on any error; trip a circuit breaker on the first hard block and *escalate to a human*, not to a workaround; collect only logged-out, and only from hosts whose terms do not prohibit it; and record the legal basis of every single quote.

6.2.2.1 The Politeness Governor

A single middleware component every fetch passes through:

1. **robots.txt resolution.** Fetch, parse (`urllib.robotparser` or `protego`), cache with TTL, snapshot to `dim_source.robots_snapshot_id`. If the target path is Disallowed for our UA, the request is **never issued** and a `ROBOTS_BLOCKED` provenance record is written. The snapshot means we can prove, months later, what the file said on the day we collected.
2. **Crawl-delay honouring.** If declared, use it. If not, default to a token bucket of 1 request per 10 seconds per host with jitter, and a daily cap.
3. **Concurrency ceiling.** At most 1 in-flight request per host. Not 8, not 4. One.
4. **Backoff.** Exponential with full jitter on 5xx and network errors; immediate stop on 403/429.
5. **Circuit breaker.** Three consecutive hard blocks on a host → open the circuit for 24 h, emit a governance alert, and mark all affected cells as `SOURCE_UNAVAILABLE` so the index suppression logic — not a silent gap — handles it.
6. **Kill switch.** A single config flag disables any host instantly, and a documented takedown contact is published in the API docs.

6.2.2.2 The Provenance Ledger

Every fetch writes an append-only record whose hash chains to its predecessor:

$$h_n = \text{SHA256}(h_{n-1} \parallel \text{canonical}(\text{record}_n)) \quad (6.1)$$

so the collection history is tamper-evident. The public `/provenance/{index_id}` endpoint returns, for any published number, the set of quotes behind it, the sources, the robots directives in force, the rate limits applied, and the chain segment. **No competing team will have this, and it is the artefact that makes the system deployable by a ministry.**

6.2.3 Why this is a stronger pitch than a CAPTCHA solver

Rehearse this exchange, because it will happen.

Judge: “The problem statement asks you to handle CAPTCHAs. You haven’t.”

You: “Correct, and deliberately. The same paragraph also requires compliance with terms of service and robots.txt, and those two requirements are mutually exclusive — a CAPTCHA is an access control, so defeating it is by definition unauthorised access under Section 43 of the IT Act. We resolved the contradiction in the direction a Government statistical product has to resolve it. Instead we built a four-rung source ladder that gets the data through licensed APIs and permitted public endpoints, and a provenance ledger that records the legal basis of every quote. Our coverage is $X\%$ of the basket; here is the coverage ledger and here is the variance

that coverage implies. A ministry can deploy this on Monday. A CAPTCHA-solving prototype cannot be deployed at all.”

6.3 Collection engine

Concern	Choice	Rationale
API fetching	httpx + tenacity	Async, HTTP/2, clean retry semantics
Static HTML at scale	Scrapy	Mature scheduler, autothrottle, middleware hooks that map onto the Politeness Governor
JS-rendered pages	Playwright (Chromium)	Real rendering; <i>used only on permitted hosts</i> . Deterministic waits on <code>network-idle</code> , never <code>sleep()</code>
Parsing	selectolax / parsel	Fast CSS/XPath; prefer embedded JSON (<code>__NEXT_DATA__</code> , JSON-LD, XHR responses) over DOM scraping — more stable and lower load
Scheduling	Celery beat (prod), APScheduler (dev)	Cron-like, distributed, observable
Idempotency	Deterministic <code>run_id</code> + <code>unique($\kappa, t, source$)</code>	Re-runs cannot double-count
Config	YAML per source, hot-reloadable	Adding a route or a source must not require a deploy

6.3.1 Collection schedule

Window	IST slot	Purpose
Primary	02:00–05:00	Lowest site load; the courteous hour. All five τ windows, full basket.
Secondary	14:00–15:00	Intraday volatility measurement on a 20% subsample
Reference data	01:00 weekly	robots.txt refresh, schedule sync, airport-charge SCD refresh
Index run	06:00	Compute, validate, publish T–1

Fixed clock time is a methodological requirement, not an operational preference — see the product key. Record the actual fetch timestamp per quote and publish the collection-time dispersion as a quality metric.

6.4 Storage and compute

- **Hot:** PostgreSQL 16 + TimescaleDB. Hypertable on `collected_at_utc`, 1-day chunks, native compression after 7 days, continuous aggregates for daily/weekly rollups. Indexes on `(route_sk, apw_days, collected_at_utc)` and on κ .
- **Cold:** Parquet on object storage, partitioned `year=/month=/route=`, queried in place by DuckDB. This is what makes the 30-day back-test re-runnable in seconds on a laptop.
- **Transformations:** SQL models in dbt-style layering (`staging` → `intermediate` → `marts`), each with tests. Keeps the index logic reviewable by a statistician who does not read Python.
- **Index engine:** a pure Python package, `apix_index`, with **no I/O**. Signature: `compute(quotes: DataFrame, weights: WeightSet, config: MethodConfig) -> IndexResult`. Purity is what makes it unit-testable against IMF Manual worked examples and what lets PSD run it on their own data without adopting our stack.
- **Numerics:** NumPy/pandas; `statsmodels` for TPD and the elasticity regression; `scikit-learn` for the Gaussian mixture; `statsmodels.tsa.x13` or STL for seasonal adjustment.

6.5 The API

Endpoint	Returns
GET /v1/index	APIx series. Params: variant (B/T/A), frequency, from, to, basis (book/travel), omega_preset, format (json/sdmx-json/csv)
GET /v1/index/routes/{route}	Route-level index with carrier decomposition
GET /v1/index/apw/{tau}	Single advance-purchase-window series
GET /v1/elasticity	Fitted $\eta(\tau)$, τ^* , coefficients, standard errors, R^2
GET /v1/components	Base / YQ / UDF / ASF / GST / convenience-fee sub-indices
GET /v1/availability	$A_g^{(t)}$ series and the matched-vs-LAF spread
GET /v1/coverage	Basket coverage ledger: routes achieved, sources live, fill rates
GET /v1/provenance/{index_id}	Full audit chain for a published number
GET /v1/methodology	Machine-readable method and weight versions in force

SDMX-JSON output is a deliberate credibility play. SDMX is the international standard for exchanging statistical data and metadata and is what statistical agencies and central banks actually consume. Emitting a valid SDMX-JSON structure — even a minimal one — says “we know how official statistics moves between institutions.” Also emit an `application/json` shape for ordinary developers, and attach an OpenAPI 3.1 spec so the docs are generated, not written.

6.6 Dashboard

View	Content
Headline	APIx daily/weekly/monthly with confidence band; y-o-y and m-o-m rates; basis toggle (booking/travel); variant toggle (B/T/A)
Sector heat map	India map or matrix, routes $\times$ time, coloured by y-o-y change. Click through to route detail
Lead-time surface	3-D or contour plot of fare index over $(\tau, \text{departure month})$; $\eta(\tau)$ curve with CI; τ^* callout per route
Component stack	Stacked area: base / YQ / UDF / ASF / GST / convenience fee, in levels and in contribution-to-change
Availability panel	$A_g^{(t)}$ area chart with the three-line matched / LAF / adjusted overlay
Back-test	APIx vs. DGCA sector averages and vs. CPI air fare item, with the agreement statistics of \$7.6
Coverage & governance	Live coverage ledger, source ladder status, circuit-breaker state, robots.txt directives in force

Build in React with Recharts (or D3 for the surface). Accessibility is not optional for a Government-facing product: WCAG 2.1 AA contrast, keyboard navigation, and **colour-blind-safe palettes** for the heat map — viridis, not red–green. English and Hindi labels via i18n. Every chart gets a “download the underlying data” button, because that is what statistical portals do.

6.7 Testing, CI, and reproducibility

Layer	What is tested
Unit — index math	Jevons/Dutot/Carli against hand-computed values; time-reversal and transitivity properties asserted; GEKS window identity; splice identities; linking-factor arithmetic against MoSPI's published 0.5267
Property-based	hypothesis: for random price vectors, assert $I^D \leq I^J \leq I^C$ where the AM–GM ordering holds; assert $I_{0,t}^J I_{t,0}^J = 1$ to floating tolerance
Unit — cleaning	Known outlier fixtures produce known dispositions
Contract — parsers	Golden-file tests against saved fixtures; a parser regression fails CI, not production
Integration	Full pipeline on a seeded database produces a byte-identical index given a fixed <code>run_id</code>
Data quality	Great Expectations suite as a blocking CI stage
Load	Collection at $3 \times$ basket size stays inside the politeness budget
Reproducibility	<code>make reproduce</code> regenerates every published figure from raw quotes and diffs against the published values

CI on GitHub Actions: lint (`ruff`), type-check (`mypy`), test with coverage gate, build images, run the reproducibility target on a fixture dataset. Everything in Docker Compose for a one-command demo.

Pin every dependency and commit the lockfile — a demo that fails to build on the judge’s laptop is a zero.

6.8 Security and governance

- Secrets in environment/vault, never in the repo; pre-commit secret scanning.
- API-key auth with per-key rate limits; read-only public tier.
- No personal data collected, stored, or inferred — state this in the README and in the API docs. It removes DPDP Act exposure and is a genuine design property, not a disclaimer.
- Row-level immutability enforced by database grants, not just by convention.
- Structured audit log of every configuration change to weights, basket, or method version, with the operator identity.
- Published takedown/contact channel, and a documented SLA for honouring a source’s request to stop.

6.9 Deployment envelope

Aspect	Target
Footprint	2 vCPU / 4 GB for collection, 2 vCPU / 8 GB for Postgres, 1 vCPU / 2 GB for API+UI
Storage growth	$\approx 4,500$ quotes/day $\times \approx 400$ B ≈ 1.8 MB/day raw; under 1 GB/year compressed
Cost posture	Runs on a single modest VM. No commercial proxy spend , which is both the compliance position and a real cost advantage
Target host	NIC / MeghRaj-class government cloud; fully containerised, no managed-service lock-in
Recovery	Nightly logical backup + Parquet cold tier means the entire index is reconstructible from raw quotes

PART 7

Unique Selling Points, In Detail

A USP is only a USP if it satisfies three tests: (i) a competing team is unlikely to have it; (ii) it cannot be retrofitted in the last four hours; (iii) it survives a hostile question. Each of the ten below is written against those tests, with an explicit build cost so the team can cut the list if time runs short. **USPs 1–5 are non-negotiable. 6–8 are strongly recommended. 9–10 are bonus.**

7.1 USP 1 — The Zero-Circumvention Architecture

Claim. APIx is the only submission that a Government of India ministry could actually deploy, because it never circumvents a technical access control and it records the legal basis of every single price quote in a tamper-evident ledger.

What it is. A four-rung source ladder (licensed APIs → partner feeds → robots-permitted public endpoints → commercial datasets), a Politeness Governor that every fetch passes through, and an append-only hash-chained Provenance Ledger exposed via a public `/provenance` endpoint.

Why others won't have it. The PS text *invites* teams to build a CAPTCHA solver and an IP rotator. Most will. They will demo it proudly, and a MoSPI evaluator — whose ministry cannot lawfully deploy such a thing — will mark it down. Reading the contradiction in the PS and resolving it deliberately is a comprehension advantage, not a technical one, which is exactly why it cannot be copied on the day.

How to demo it. Live: hit a host whose `robots.txt` disallows the fare path. Show the request *not being issued* and a `ROBOTS_BLOCKED` ledger entry appearing instead. Then call `/provenance/{id}` on a published index number and walk the chain back to the source, the directive, and the rate limit that applied. Thirty seconds; nobody else will have anything like it.

Build cost. ≈6 hours. Politeness Governor 3 h, ledger + hash chain 2 h, provenance endpoint 1 h.

The hostile question. “*So you have less data.*” Answer: yes, and here is the coverage ledger quantifying exactly how much less, and the index standard errors that implies. Then: “*inadmissible data has a coverage of zero, because it cannot be published.*”

7.2 USP 2 — CPI-2024-Native by Construction

Claim. APIx is not “compatible with” the CPI — it is built on the CPI 2024 series’ own mathematics, classification and linking arithmetic. Integration is a configuration change, not a migration project.

What it is. Jevons elementary aggregates and Young/Modified Laspeyres higher-level aggregation, matching MoSPI FAQ Q.20 and Q.21 exactly. COICOP 2018 sub-class 07.3.3 tagging throughout. A linking-factor implementation validated against MoSPI’s published General-level factors (Rural 0.5222 / Urban 0.5320 / Combined 0.5267). A weight vector that is an *injected parameter*, so the Price Statistics Division supplies routes and weights and the module consumes them — which is a literal reading of the PS phrase “*an index-construction module based on PSD given routes and weights.*”

Why others won't have it. It requires reading the CPI 2024 FAQ and the Expert Group Report. Most teams will use a mean, or a Laspeyres because it is the one they remember from class, and will not

know that MoSPI switched to COICOP 2018 in February 2026.

How to demo it. A single toggle on the dashboard: “APIx own base” $\leftrightarrow$ “rebased to CPI 2024=100 via linking factor”, with the official air fare item index overlaid. Say out loud: *“Jevons at the bottom, Young at the top — because that is what you do.”*

Build cost. $\approx$ 4 hours, mostly in the index engine, which you are building anyway.

The hostile question. “Why Jevons and not Dutot?” Answer with the AM–GM worked example from §3.2, the time-reversal property, and the lognormality of fares. Have the three-formula slide ready.

7.3 USP 3 — The Dual-Time-Basis Index

Claim. APIx publishes both the *acquisition-basis* index (what it costs to buy a flight today) and the *use-basis* index (the price of travel consumed this month), and reconciles them. No Indian price series distinguishes the two.

What it is. §4.7. The booking-basis series attributes a quote to the collection date; the travel-basis series attributes it to the departure date. The UK ONS records air fare changes *in the month the flight departs, not when the ticket is bought* — so the travel basis is the internationally CPI-consistent one, while the booking basis is what a consumer, a regulator or an RBI nowcaster actually wants.

Why others won’t have it. Almost nobody knows the distinction exists. It comes directly from reading foreign NSI methodology, and it is the clearest possible signal that the team engaged with price-statistics literature rather than with scraping tutorials.

How to demo it. One chart, two lines, a basis toggle, and one sentence: *“These are the same quotes. This one answers ‘what does a ticket cost today’; this one answers ‘what did air travel cost this month’. CPI needs the second. RBI wants the first. We publish both.”*

Build cost. $\approx$ 2 hours. It is a re-attribution of the same fact table — a different `GROUP BY`, essentially free once the schema stores both t and d .

The hostile question. “Which one is the headline?” Answer: travel basis is the headline because it is CPI-consistent; booking basis is published alongside and is the one used for the nowcast.

7.4 USP 4 — The Availability-Adjusted Index

Claim. When cheap fare buckets sell out, a naive matched index reports *zero inflation* during exactly the surge episodes consumers complain about. APIx detects this, quantifies it, and corrects it with a continuous, monotone adjustment that reduces to standard practice in the normal case.

What it is. §4.8. The availability ratio $A_g^{(t)}$, a lowest-available-fare index, and the blend $I^{\text{adj}} = (I^{\text{matched}})^A (I^{\text{LAF}})^{1-A}$.

Why others won’t have it. It requires noticing that missingness in fare data is *informative* — a selection-on-the-dependent-variable problem. Teams that treat a sold-out flight as a failed scrape will silently build the bias in and never see it.

How to demo it. The strongest ninety seconds in the deck. Pick a real surge period. Three lines: matched (flat, wrong), LAF (spiking), adjusted (spiking, correct). Underneath, $A_g^{(t)}$ as an area chart collapsing toward zero. Then: *“This is the Kumbh problem and the festival-fare problem. A naive index cannot see it. Ours can.”*

Build cost. ≈ 5 hours: availability capture in the parser (2 h), LAF index (1 h), blend and charting (2 h).

The hostile question. “*Is that blend principled or invented?*” Answer honestly: it is a proposed estimator, not an established standard. Defend it on three properties — it is continuous in A , monotone, and exactly equals the matched index when $A = 1$, so it never disturbs the normal case. Then note that the alternative (dropping sold-out cells) is *also* not an established standard, and is provably biased. Offer the LAF series as a published diagnostic either way. **Do not claim it is official methodology.**

7.5 USP 5 — The Lead-Time Elasticity Surface

Claim. APIx treats advance-purchase window as a first-class published dimension, estimates the fare-lead-time elasticity $\eta(\tau)$ with standard errors, and reports the optimal booking lead time τ^* per route.

What it is. The pooled fixed-effects log-linear model of §4.9, the elasticity $\eta(\tau) = \beta_1 + 2\beta_2 \ln \tau$, the trough $\tau^* = \exp(-\beta_1/2\beta_2)$, and the Gaussian-mixture recovery of the underlying fare ladder.

Why others won’t have it. The PS asks for “lead-time elasticity curves” and most teams will plot average fare against τ and call it an elasticity curve. An average is not an elasticity. Running an actual regression with route/carrier/day-of-week/month fixed effects and clustered standard errors takes half an hour and is a completely different object.

How to demo it. The 3-D surface is the visual, but the *number* is the payload: “*For DEL–BOM, τ^* is 26 days — book about four weeks out. Elasticity at one week is -0.34 : a 10% longer lead time buys you 3.4% off.*” That is a press headline, and press headlines are what MoCA and DGCA want from a transparency product.

Build cost. ≈ 4 hours. `statsmodels` regression 1 h, surface plot 2 h, mixture model 1 h.

The hostile question. “*Is that causal?*” Answer: no — it is a conditional association within a revenue-management system, and the fixed effects absorb route, carrier, day-of-week and seasonal confounders but not unobserved demand. Say “association”, never “causal effect”. Knowing the difference is itself the answer.

7.6 USP 6 — The Rigorous Back-Test Harness

Claim. The PS requires 30 days of back-tested results against DGCA fare data. APIx delivers that as a *statistical agreement analysis*, not an eyeballed overlay.

What it is. A harness that compares the APIx monthly movement against (a) DGCA sector-average fare movements and (b) the CPI 2024 air fare item index, and reports:

Statistic	What it answers
Pearson and Spearman ρ	Do they move together, in level and in rank?
Mean absolute deviation of the y-o-y rate	By how much do we differ, in inflation points?
Bland–Altman plot	Is the disagreement constant, or does it grow with the level? (The right tool for method-agreement, and almost nobody uses it)
Directional agreement rate	In what % of months do both move the same way? Often the most persuasive single number
Theil’s U vs. random walk	Does APIx beat a naive forecast?
Diebold–Mariano test	Is the forecast improvement statistically significant?

Why others won’t have it. Most will show two lines and say “as you can see, they track closely.” A

Bland–Altman plot and a Diebold–Mariano statistic is a different register of seriousness.

How to demo it. Show the overlay, then say: “*and here is the agreement analysis*” and put up the statistics table. Report an unflattering number if you have one; it buys enormous credibility.

Build cost. ≈4 hours.

The 30-day problem, and how to be honest about it. If collection starts at the hackathon, there is no 30-day history. Run **two tracks** and label them unambiguously:

- **Track A (live).** Start collecting the moment the basket is defined — ideally weeks before the finale. Even 14 real days is worth more than 90 synthetic ones.
- **Track B (replay).** A synthetic generator calibrated to published DGCA monthly average-fare movements and to known seasonality, used to exercise the pipeline at scale.

Put a visible **SYNTHETIC** watermark on every Track B chart. If a judge discovers an unlabelled synthetic chart, the entire submission’s credibility is gone. If you label it, you look disciplined.

7.7 USP 7 — The CPI Transport Nowcast

Claim. APIx is available daily; the CPI air fare item arrives monthly around the 12th. APIx therefore *leads* the official series, and we quantify by how much.

What it is. The mixed-frequency bridge regression of §4.10, with an ATF control from PPAC, evaluated out-of-sample against a random-walk benchmark, plus Granger-causality and lead–lag cross-correlation.

Why others won’t have it. It requires connecting the airfare work to the actual monetary-policy use case. This is the USP that speaks to RBI rather than to MoSPI, and it turns “a dashboard” into “an economic indicator”.

How to demo it. A chart with the official monthly series as step points and the APIx daily series continuous, with the nowcast for the not-yet-published month highlighted and its confidence band shown.

Build cost. ≈3 hours — but it is **blocked on having history**. If Track A is short, present the methodology and the code, and label the result as illustrative. Do not fabricate an RMSE.

The hostile question. “*Air fare is a tiny weight. Who cares?*” Answer: agreed on the weight — Transport is 8.796% of combined CPI and air fare is a slice of that. The claim is not that this moves headline inflation. The claim is that the *method* generalises: the same pipeline nowcasts hotel tariffs, cab fares, and the twelve online markets MoSPI just introduced. Air fare is the pilot, not the prize.

7.8 USP 8 — Hash-Chained Reproducibility

Claim. Every published APIx number can be regenerated from raw quotes by a single command, and the collection history is tamper-evident.

What it is. `method_version` and `weights_version` stamped on every index value; a `make reproduce` target that recomputes the entire published series from the Parquet cold tier and diffs against `fact_index_value`; the hash-chained provenance ledger.

Why others won’t have it. Reproducibility is invisible until someone asks for it, so nobody builds it under time pressure. But it is a first-order requirement for official statistics — an index number that cannot be audited cannot be published by a ministry.

How to demo it. Run `make reproduce live`. It prints 0 differences across N index values. Then flip one byte in a stored quote and re-run: the hash chain breaks and the harness reports exactly which record and which downstream index values are affected.

Build cost. ≈ 3 hours if the schema was designed for it from the start; essentially impossible to bolt on afterwards. **Decide this in hour one.**

7.9 USP 9 — The Frugal Deployment Envelope

Claim. APIx runs on a single modest VM with zero commercial proxy spend, and is deployable on NIC/MeghRaj-class government infrastructure.

What it is. The costing in §6.9. Roughly 1.8 MB of raw quotes per day, under a GB per year compressed, one small VM for collection and one for the database.

Why it matters. A scraping-first architecture needs a residential proxy pool, which costs real money *per gigabyte* and is a recurring line item a ministry must justify. It is also the exact expenditure that makes a compliance reviewer uncomfortable. The compliance choice and the cost choice are the same choice — point that out; it is a nice piece of narrative economy.

How to demo it. A single slide: monthly cost, storage growth curve, and the statement “*no external paid dependency in the critical path.*”

Build cost. ≈ 1 hour to instrument and chart.

7.10 USP 10 — The Fare Transparency Layer

Claim. APIx implements, as a working prototype, two of the three recommendations the Parliamentary Standing Committee on Transport, Tourism and Culture made to the Ministry of Civil Aviation in its 375th Report.

What it is. The Committee, reporting on Demands for Grants 2025–26, found the DGCA *lacks capacity to proactively monitor airfare trends* and recommended (i) an AI-driven predictive fare monitoring system, (ii) a public “**Pricing Transparency Index**” rating airlines on adherence to fair pricing, and (iii) an “**Airfare Vigil**” consumer reporting app. APIx’s route-level index, elasticity surface and fare-ladder recovery are (i); a per-carrier tariff-band compliance score against Rule 135 of the Aircraft Rules, 1937 is (ii).

Why others won’t have it. It requires knowing the report exists. This is a research advantage, and it lets the pitch end on “*this is not a hackathon idea; this is a standing Parliamentary recommendation, prototyped.*”

How to demo it. A consumer-facing panel: for each carrier and route, the observed fare distribution against the declared tariff band, an outlier count, and a transparency score. Frame it as *additional* to the CPI mandate, not a substitute — MoSPI owns the PS, and the CPI work is the main line.

Build cost. ≈ 3 hours, and it depends on obtaining declared tariff bands, which may not be publicly available. **If the bands are not obtainable, build the panel on observed dispersion instead and say so.**

7.11 USP summary: build order under time pressure

#	USP	Hours	Priority	Blocked on
1	Zero-Circumvention Architecture	6	Must	Nothing
2	CPI-2024-Native	4	Must	Weight data
3	Dual-Time-Basis	2	Must	Schema decision
4	Availability-Adjusted Index	5	Must	Availability capture in parser
5	Lead-Time Elasticity Surface	4	Must	≥2 weeks of data
8	Hash-Chained Reproducibility	3	High	Hour-one schema decision
6	Back-Test Harness	4	High	DGCA history
9	Frugal Deployment Envelope	1	High	Instrumentation
7	CPI Transport Nowcast	3	Medium	History length
10	Fare Transparency Layer	3	Bonus	Tariff bands

Total for the five “Must” USPs: **21 hours**, on top of the base build. That is the realistic scope. Cutting 7 and 10 costs nothing in the pitch; cutting 1 or 4 costs the whole differentiation.

PART 8

Delivery: Plan, Risks, and Defence

8.1 What the PS explicitly demands, and where each item is satisfied

The “Expected Solution” paragraph is a checklist. Judges will score against it literally. Map every clause.

PS requirement	Section	Status / note
Multi-source scraping engine in Python (Scrapy/Selenium/Playwright), scheduled daily	§6.3	Scrapy + Playwright + httpx behind the Source Ladder. Selenium deliberately not used (Playwright supersedes it) — say so
Ethically designed	§6.2	The Zero-Circumvention Policy is the answer
Handles JS rendering	§6.3	Playwright on permitted hosts
Handles CAPTCHAs / anti-bot / IP rotation	§6.2	Deliberately not implemented. Rehearse the answer in §6.2.3 — this is a scored risk and must be addressed head-on, not skipped
Cleaned, de-duplicated database with full metadata	§5.3	Star schema; the PS’s listed fields are all present, plus availability and provenance
Separates base fare from taxes, UDF, convenience charges	§4.2.1	Full decomposition with SCD tables for administered charges
Removes outliers, handles missing values, cancellations/sold-out	§4.2.2–4.2.3, §4.8	Tukey + Hampel; three-case imputation; availability adjustment goes beyond the ask
Index module based on PSD-given routes and weights	§4.5	Weight vector is an injected parameter; PSD supplies it
Daily, weekly, monthly frequencies	§4.11	Daily; weekly 7-day GM; monthly calendar GM
Multiple advance-purchase windows	§4.1	τ is a dimension of the product key, not a filter
T+1/7/15/30/45		
Basket of representative city-pairs on DGCA traffic	§5.2	Derived, stratified, and shown to reproduce the PS’s named sectors
Interactive dashboard: trends, sector heat maps, lead-time elasticity curves	§6.6	All three, plus availability and coverage panels
API consumable by NSO and RBI	§6.5	REST + SDMX-JSON
Documentation	—	This document + README + OpenAPI + methodology note
Automated testing	§6.7	Unit, property-based, contract, integration, data-quality, reproducibility
≥30 days back-tested vs. DGCA monthly average fares	§7.6	Two-track plan; label synthetic data SYNTHETIC

8.2 The 30-day back-test plan

When	Action	Output
D-45	Freeze basket v1; stand up collection on rung-1 sources only	Live quotes accumulating
D-40	Add rung-3 sources that pass the robots/ToS gate	Coverage rises; ledger records basis
D-30	Back-test clock starts	The 30-day window the PS asks for
D-21	First elasticity fit; sanity-check τ^* against intuition	Model validated
D-14	First DGCA monthly release lands inside the window; run agreement stats	Bland-Altman, ρ , directional agreement
D-7	Freeze method version; run <code>make reproduce</code>	Reproducibility report
D-3	Generate the final evidence pack	Charts, tables, statistics
D-0	Demo	Live collection running on stage

If you take one operational instruction from this document, take this one: start collecting **now**. Not at the hackathon. The single highest-value, lowest-effort action available is a cron job on the cheapest VPS you can rent pulling ten routes from a free-tier fare API into a SQLite file. Every day you delay costs a day of the only asset that cannot be built under time pressure. A team with 45 days of real fare history and a mediocre dashboard beats a team with a beautiful dashboard and four hours of data. Every time.

8.3 Team allocation (six members)

Role	Owns	Hard deliverable
Data engineer	Source Ladder, Politeness Governor, parsers, scheduler	Quotes landing reliably; ledger populated
Statistician	Index engine, cleaning, elasticity, back-test	<code>apix_index</code> package with tests passing against IMF worked examples
Backend	Schema, API, SDMX output, provenance endpoint	OpenAPI spec live, <code>/provenance</code> working
Frontend	Dashboard, surface plot, heat map, accessibility	The three-line availability chart and the elasticity surface
DevOps/QA	Docker, CI, Great Expectations, <code>make reproduce</code>	One-command demo that works on a cold laptop
Domain/pitch	CPI & DGCA research, methodology note, deck, Q&A drill	This document turned into 8 slides and 6 rehearsed answers

Two rules. (1) The statistician and the data engineer agree the schema in hour one — reproducibility and dual-basis both depend on schema decisions that cannot be retrofitted. (2) The pitch owner is not also building a feature in the last six hours.

8.4 Risk register

Risk	Prob.	Impact	Mitigation
Free-tier API quota exhausted mid-demo	Med	High	Cache aggressively; demo runs off stored data with one live call for theatre; have a second provider keyed
Parser breaks on a site redesign	High	Med	Golden-file contract tests fail in CI, not on stage; parsers are config, not code; degrade to fewer sources rather than crash
Host blocks collection	Med	Med	Circuit breaker; index suppression handles it <i>as designed</i> ; the coverage ledger shows it honestly — turn it into a demo of governance
Insufficient history for elasticity/nowcast	High	High	Start collecting at D–45. If short: present method + code, label results illustrative, never fabricate
Judge objects to no-CAPTCHA	Med	High	Rehearsed answer, §6.2.3. Frame as comprehension, not limitation
Route weights unavailable at sector level	Med	Med	Fall back to airport-pair proxies from AAI airport traffic; document the approximation
Booking lead-time distribution unknown	High	Med	Three declared presets + tornado sensitivity chart. Disclose, don't hide
Demo laptop/network failure	Med	High	Everything runs offline from a seeded database; record a 90-second backup video
Scope creep	High	High	USP table §7.11 is the cut list. Five Musts, then stop
Team quotes a stale number on stage	Med	Med	Every [VERIFY] in this document re-pulled in the final week and written on one card

8.5 The judge Q&A drill

Rehearse these out loud. Each answer is 30–45 seconds.

1. “MoSPI already collects airfares online. What are you adding?”

Daily instead of weekly; route-level instead of twelve online markets; five advance-purchase windows instead of none; availability adjustment; component decomposition; a public API. The delta table in §1.2. Lead with the fact that you know CPI 2024 already does online airfare collection — that is the credibility moment.

2. “You didn’t do CAPTCHAs.”

§6.2.3, verbatim. The PS is internally contradictory; we resolved it the way a ministry must.

3. “Why Jevons?”

Because MoSPI uses it (FAQ Q.20); because it satisfies time reversal and transitivity; because fares are lognormal. Show the three-formula slide.

4. “How do you know your index is right?”

We don’t “know”; we bound it. Standard errors by delta method, flight-block bootstrap CIs, GEKS-Jevons and TPD as independent robustness variants with the spread published, and agreement statistics against DGCA. **Never say “accurate”; say “bounded and reproducible”.**

5. “Is this legal?”

Walk the four instruments (IT Act s.43/66, Copyright Act, Contract Act, DPDP), state that **no personal data is collected** so DPDP does not attach, then show the provenance ledger.

6. “What happens when a site changes its layout?”

Contract tests fail in CI. The source degrades to unavailable, the index suppresses affected cells, coverage drops visibly. The system fails loudly and correctly, not silently and wrongly.

7. “How is this different from a flight price comparison site?”

A comparison site answers “what should I pay for this trip.” A price index answers “how has the price of air travel changed.” Different objects: one needs the cheapest current offer, the other needs a constant-quality matched basket, weights, and a formula. That is why the product key exists.

8. “Can this scale beyond airfares?”

Yes, and that is the real pitch to DIID. The Source Ladder, provenance ledger, cleaning pipeline and index engine are item-agnostic. Airfare is the hardest case — dynamic pricing, perishable

inventory, availability censoring. Hotels, cabs and the twelve online markets are strictly easier. **This is the answer that turns a project into a platform, and it is the strongest closing line available.**

8.6 The eight-slide deck

Slide	Title	The single thing it must land
1	The reframe	CPI 2024 already collects airfares online. Here is what it still cannot do.
2	The contradiction	The PS asks for CAPTCHA-defeating <i>and</i> ToS compliance. We chose compliance. Here is the ladder.
3	The method	Jevons at the bottom, Young at the top — because that is what MoSPI does. Three-formula worked example.
4	Availability	Three lines and a collapsing area chart. The surge problem, seen and fixed.
5	Lead time	The surface, $\eta(\tau)$, and τ^* per route.
6	Evidence	Back-test agreement statistics. Include one unflattering number.
7	Governance	Live <code>/provenance</code> call + <code>make reproduce</code> printing zero diffs.
8	The platform	Airfare is the hardest item. The pipeline generalises to every online market MoSPI just added.

8.7 Deliverables checklist

Artefact	Acceptance criterion
Source repository	<code>docker compose</code> up works on a cold machine; README under 2 pages to first index number
<code>apix_index</code> package	Pure, typed, >85% coverage, IMF worked examples as tests
Seeded database dump	Demo runs fully offline
OpenAPI 3.1 spec	Validates; <code>/provenance</code> and <code>/coverage</code> present
Dashboard build	Static bundle; WCAG AA contrast; data download on every chart
Methodology note	6–8 pages in MoSPI discussion-paper format; every formula numbered
Evidence pack	Back-test charts + agreement statistics; synthetic clearly watermarked
Reproducibility report	<code>make reproduce</code> output showing zero differences
Coverage ledger	Routes achieved, sources live, legal basis per source, fill rates
Deck	8 slides, no slide with more than 20 words
Backup video	90 seconds, shows the four live moments

8.8 Closing assessment

An honest read on this problem statement. It is a good PS for a team that can do statistics and a bad one for a team that only writes scrapers — which is why it is worth picking if, and only if, at least one member is willing to own the index-number theory. The scraping is the easy, commoditised half; every team will have it, and several will have more coverage than you. The differentiation is entirely in three places: **knowing that CPI 2024 already exists and already collects airfares online; resolving the PS’s internal contradiction deliberately rather than ignoring it; and treating sold-out inventory as a price signal rather than as missing data.** Those three things cost roughly a day of reading and eleven hours of building, and no amount of dashboard polish substitutes for them. Also: this PS is from MoSPI, which means the evaluator is likely a statistician, not a software engineer. Optimise the pitch accordingly. A correct standard error will land better than an animated map.

PART 9

Sources and Further Reading

Every factual claim in this document traces to one of the entries below. URLs verified at time of writing; re-check before quoting on stage.

9.1 Primary — Indian official statistics

- [1] MoSPI, *Frequently Asked Questions on CPI 2024 Series* (Annexure V to the CPI press release of 12 February 2026). **The single most important source for this project.** Establishes: base 2024=100; HCES 2023–24 weights; COICOP 2018 with 12 divisions / 43 groups / 92 classes / 162 sub-classes; 358 weighted items; 1,465 rural and 1,395 urban markets across 434 towns plus 12 online markets; **Jevons** for elementary and **Young/Modified Laspeyres** for higher-level indices; weekly online price collection; **“Airtfares are collected through well-known online platforms”**; division-wise weights including Transport at 8.796% combined; the linking-factor formula and the General-level factors 0.5222 / 0.5320 / 0.5267.
https://www.mospi.gov.in/uploads/documents/documents/1770891066052-Annexure_V.pdf
- [2] MoSPI, *Press Release: Consumer Price Index on base 2024=100 for January 2026* (12 February 2026). First release of the new series; 2.75% y-o-y.
https://www.mospi.gov.in/uploads/latestReleases/latest_release_1770891893893_6b458c0a-c327-4fef-a554-41131ea67273_Press_Release_of_CPI_for_Jan26.pdf
- [3] MoSPI, *Expert Group Report on Comprehensive Updation of the Consumer Price Index*. The methodology of record: COICOP mapping, the role of the **Price Statistics Division (PSD)**, the introduction of online markets, and the list of items collected via alternative data sources. Note the HCES 2023–24 finding that online platforms account for 4.0% of rural and 10.5% of urban household expenditure — the justification for the online-market stratum.
<https://www.mospi.gov.in/uploads/Marquee/doc-0aee0026-bf32-4213-ae4f-c5308088444e.pdf>
- [4] MoSPI, *Press note on release of the Expert Group Report*.
<https://cpi.mospi.gov.in/PDFile/Press%20note%20on%20release%20of%20Expert%20Group%20Report%20on%20Comprehensive%20Updation%20of%20CPI.pdf>
- [5] MoSPI, CPI product page and monthly releases.
<https://www.mospi.gov.in/themes/product/9-consumer-price-index-cpi>
- [6] MoSPI, *e-Sankhyiki* macro-indicators portal — the nodal dataset named in the problem statement.
<https://esankhyiki.mospi.gov.in/macroindicators?product=cpi>
- [7] MoSPI, CPI portal (item-wise weights under Announcements). <https://cpi.mospi.gov.in>
- [8] MoSPI, *Press Release: CPI for December 2025* — the final release on base 2012=100, confirming the changeover date.
<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2213736>
- [9] MoSPI, *Discussion Paper on the proposed housing index compilation methodology* (October 2025). Useful as a **template for the format and register of the methodology note** APIx should submit.

9.2 Primary — Indian civil aviation

- [10] DGCA, monthly *Traffic Report* (domestic). Passengers carried, market share, passenger load factor, cancellations and causes, complaints, on-time performance. Establishes: 864.04 lakh passengers Jan–Jun 2026 (+1.44% y-o-y); 984.03 lakh Jan–Jul 2026; IndiGo 66.3% (June) and 67.4% (July) domestic share; Air India Group ≈24%; Akasa 5.5%; SpiceJet 1.6%; PLFs Akasa 92.2% / SpiceJet

87.8% / AI Group 85.5% / IndiGo 85.1%; cancellation rate 0.63%.

<https://www.dgca.gov.in>

- [11] Rajya Sabha, reply of the Minister of State for Civil Aviation, 8 December 2025. Establishes the **Tariff Monitoring Unit**: set up 2010, monitors **78 selected routes**, *monthly, manually, using airlines’ websites*, covering **≈27% of domestic traffic**; airfares remain deregulated under **Rule 135 of the Aircraft Rules, 1937**.
- [12] Ministry of Civil Aviation, reply to Parliament, July 2026. Establishes a **20.5% increase in average airfare across 72 domestic sectors** comparing June 2026 with March 2025, and that **ATF is 35–40% of airline operating cost** with prices up over 100% on the West Asia conflict.
- [13] Parliamentary Standing Committee on Transport, Tourism and Culture, *375th Report: Demands for Grants (2025–26) of the Ministry of Civil Aviation*. Finds the DGCA lacks capacity to proactively monitor airfare trends; recommends an AI-driven predictive monitoring system (“AirPrice Guardian”), a public “**Pricing Transparency Index**”, and an “**Airfare Vigil**” consumer app. **The basis of USP 10**.
- [14] Reporting on the DGCA’s December 2024 request to airlines for two years of ticket-level data (booking date, fare charged, base fare) and the Federation of Indian Airlines’ refusal on commercial-confidentiality grounds (letter of 11 March 2025). **The evidence that the booking lead-time distribution is genuinely unobtainable**.
- [15] AERA tariff orders (User Development Fee). <https://aera.gov.in>
- [16] Airports Authority of India traffic statistics. <https://www.aai.aero>
- [17] PPAC, Ministry of Petroleum & Natural Gas — ATF and fuel prices. <https://ppac.gov.in>

9.3 International methodology — index numbers and alternative data

- [18] IMF/ILO/OECD/UNECE/Eurostat/World Bank, *Consumer Price Index Manual: Concepts and Methods* (2020). The canonical reference. Elementary formulae, axiomatic tests, Laspeyres/Paasche/Fisher/Törnqvist, Young and Lowe. <https://www.imf.org>
- [19] W. E. Diewert, *Scanner Data, Elementary Price Indexes and the Chain Drift Problem* — companion chapter to the CPI Manual. Multilateral indices (GEKS, GEKS-Törnqvist / CCDI), Time Product Dummy and Weighted TPD, missing-product handling. Explicitly cites **airlines and hotels as the dynamic-pricing case** motivating multilateral methods.
<https://www.imf.org/~media/Files/Data/CPI/companion-publication/chapter-6-chain-drift-problem-and-multilateral-indices.ashx>
- [20] Eurostat, *Guidelines on the use of web scraped data in the HICP* (November 2020). Establishes that weights are typically unavailable for web-scraped data, that fixed-base Jevons is the standard approach, that **GEKS-Jevons and Time Product Dummy** are the multilateral options, and that Dutot implicitly weights by base-period price level.
<https://ec.europa.eu/eurostat/documents/272892/12032198/Guidelines-web-scraping-HICP-11-2020.pdf>
- [21] ONS, *Aggregate index of air fares methodology*. **The single best international precedent for this project**. Establishes ONS’s advance-collection windows (long-haul 6/3/1 months; short-haul 3/1; domestic 1 month), collection on a commercially confidential “index day” (usually the 2nd or 3rd Tuesday), inclusion of return legs, and — critically — that **air fare changes are recorded in the CPI in the month the flight departs, not when the ticket is bought. The basis of USP 3**.
<https://www.ons.gov.uk/aboutus/transparencyandgovernance/freedomofinformationfoi/aggregateindexofairfaresmethodology>
- [22] ONS, *Domestic, European and long-haul airfares consumer prices sub-indices* (ad hoc release, Jan 2017 – Feb 2025). Shows a national statistical institute publishing airfare sub-indices by *advance-purchase window* — direct precedent for the APIx τ dimension.
<https://www.ons.gov.uk/economy/inflationandpriceindices/adhocs/2716domesticeuropeana ndlonghaulairfaresconsumerpricessubindicesjanuary2017tofebruary2025>

- [23] ONS, *New index number methods in consumer price statistics* (2020). Scores methods against a framework; finds Quality-adjusted Geary–Khamis best where expenditure information exists and recommends **GEKS-Jevons in its absence** — i.e. the APIx case.
<https://www.ons.gov.uk/economy/inflationandpriceindices/articles/newindexnumbermethodsinconsumerpricestatistics/2020-09-01>
- [24] ONS, *Introducing multilateral index methods into consumer price statistics*. GEKS-Törnqvist with a 25-month window and a **mean splice**; the splicing taxonomy.
<https://www.ons.gov.uk/economy/inflationandpriceindices/methodologies/introducingmultilateralindexmethodsintoconsumerpricestatistics>
- [25] ONS, *How multilateral index methods help us understand grocery scanner data* (2026). Confirms GEKS-Törnqvist already in use for rail fares and second-hand cars, and adopted for grocery scanner data from March 2026.
<https://www.ons.gov.uk/economy/inflationandpriceindices/methodologies/howmultilateralindexmethodshelpusunderstandgroceryscannerdata>
- [26] ONS, *Research indices using web scraped price data* (2017 update). Bilateral vs. multilateral taxonomy; the CLIP index; GEKS sensitivity to atypical prices.
- [27] K. Van Loon (Statbel), *Scanner data and web scraping in the Belgian CPI* (2020). Practical implementation notes; GEKS-Törnqvist, 25-month rolling window, half splice.
- [28] National Academies of Sciences, Engineering, and Medicine, *Modernizing the Consumer Price Index for the 21st Century*. Chain drift, multilateral methods, and the concordance problem between non-survey data hierarchies and CPI nomenclature.
<https://www.nationalacademies.org/read/26485/chapter/4>
- [29] J. Lent and A. H. Dorfman, *A transaction price index for air travel*, *Monthly Labor Review*, BLS, June 2005. Why listed fares differ from transaction prices, and the imputation and matching problems specific to air travel.
<https://www.bls.gov/opub/mlr/2005/06/art2full.pdf>
- [30] BLS, *Import and Export Air Passenger Fares Price Indexes* fact sheet.
<https://www.bls.gov/mxp/publications/factsheets/air-passenger-facts.pdf>
- [31] BLS, CPI for All Urban Consumers: Airline Fares (series CUSR0000SETG01), via FRED — a 1989–2026 comparator series.
<https://fred.stlouisfed.org/series/CUSR0000SETG01>
- [32] UN Statistics Division, *Classification of Individual Consumption According to Purpose (COICOP) 2018*. The classification MoSPI adopted for CPI 2024; source of the 07.3.3 code.

9.4 Legal — data collection in India

- [33] *Information Technology Act, 2000*, ss. 43 and 66 — compensation and criminal liability for accessing, downloading, copying or extracting data from a computer resource without the owner’s permission; s. 43 also covers disruption.
- [34] *Copyright Act, 1957*, s. 13(1)(a); *Eastern Book Company v. D.B. Modak* (Supreme Court of India) — the “skill and judgment” standard under which a compilation attracts copyright.
- [35] *Indian Contract Act, 1872* — terms of use as an enforceable contract.
- [36] *Digital Personal Data Protection Act, 2023* — engages only where personal data is processed. **Not applicable to APIx**, which collects no personal data.
- [37] Reply of the Minister of State for Electronics and IT, Rajya Sabha (August 2024) — the Government’s position that scraping publicly available data may breach s. 43 of the IT Act.
- [38] *hiQ Labs v. LinkedIn* (US 9th Cir., 2022) — scraping public pages is not “unauthorised access” under the CFAA.
- [39] *Meta v. Bright Data* (2024) — terms bind account holders; **logged-out** public scraping is materially more defensible. Directly motivates the unauthenticated-session requirement in the APIx product

key.

- [40] *eBay v. Bidder's Edge* (N.D. Cal., 2000) — trespass to chattels at extreme request volumes; the origin of the rate-limiting-as-legal-defence argument.

9.5 Standards and tooling

- [41] SDMX (Statistical Data and Metadata eXchange) — the ISO standard for statistical data exchange between agencies. <https://sdmx.org>
- [42] Robots Exclusion Protocol, RFC 9309.
- [43] Scrapy, Playwright, `protego`, `httpx`, TimescaleDB, DuckDB, Great Expectations, `statsmodels`, `hypothesis` — project documentation.

Reading order if you only have four hours. [1] the CPI 2024 FAQ, cover to cover (20 min — it is seven pages and it is the whole reframe). [21] the ONS air fares methodology page (10 min — it gives you USP 3). [20] the Eurostat web-scraping guidelines, the index-formula section (30 min — it gives you USP 2's defence). [13] the Parliamentary Standing Committee report summary (15 min — it gives you USP 10). [19] Diewert on chain drift, skim (45 min — it gives you the answer to “what about multilateral methods?”). Then spend the remaining time on [10], the DGCA monthly traffic report, and build your route weights.